Intro to Neuroscience: Behavioral Neuroscience

# Animal Navigation II: Brain mechanisms

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# Outline of the two lectures on Navigation

- **Introduction: Feats of animal navigation**
- **Navigational strategies:**
  - **Beaconing**
  - **Route following**
  - **Path integration**
  - **Map and Compass / Cognitive Map**
- **Sensory cues for navigation:**
  - **Compass mechanisms**
  - **Map mechanisms**
- **Brain mechanisms of Navigation (brief introduction)**
- **Summary**

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# Brain Mechanisms of Navigation – Outline

- **Hippocampus and spatial memory: early discoveries**
- **Hippocampus and large-scale navigation**
- **Back to small-scale navigation in the laboratory:**
  - **Place cells**
  - **Head direction cells**
  - **Grid cells**
- **Toward a neurobiology of large-scale navigation**
- **Other brain areas involved in navigation**



# The 2014 Nobel Prize in Physiology or Medicine



**John O'Keefe**
Born 1939, USA
University College London



**May-Britt Moser**
Born 1963, Norway
Norwegian University of Science and Technology, Trondheim



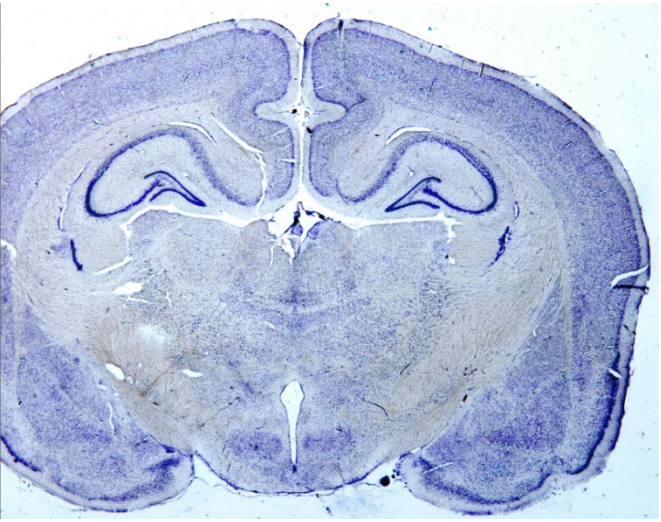
**Edvard I. Moser**
Born 1962, Norway
Norwegian University of Science and Technology, Trondheim

# Brain Mechanisms of Navigation – Outline

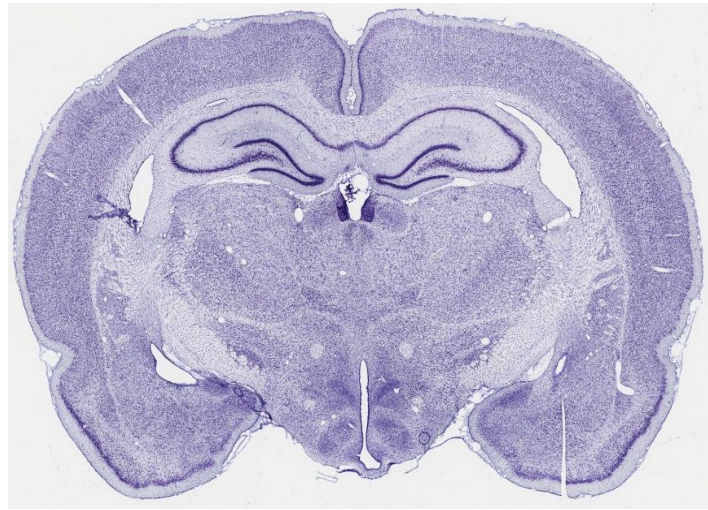
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  - **Other brain areas involved in navigation**

# The hippocampus

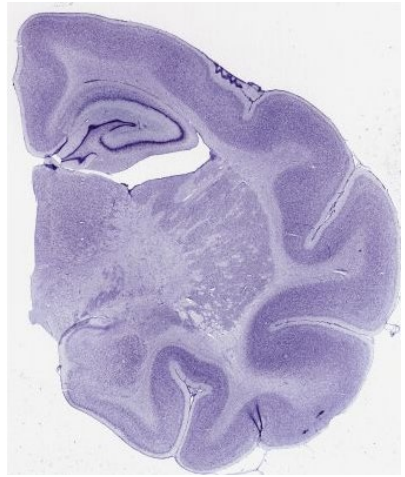
Egyptian fruit bat



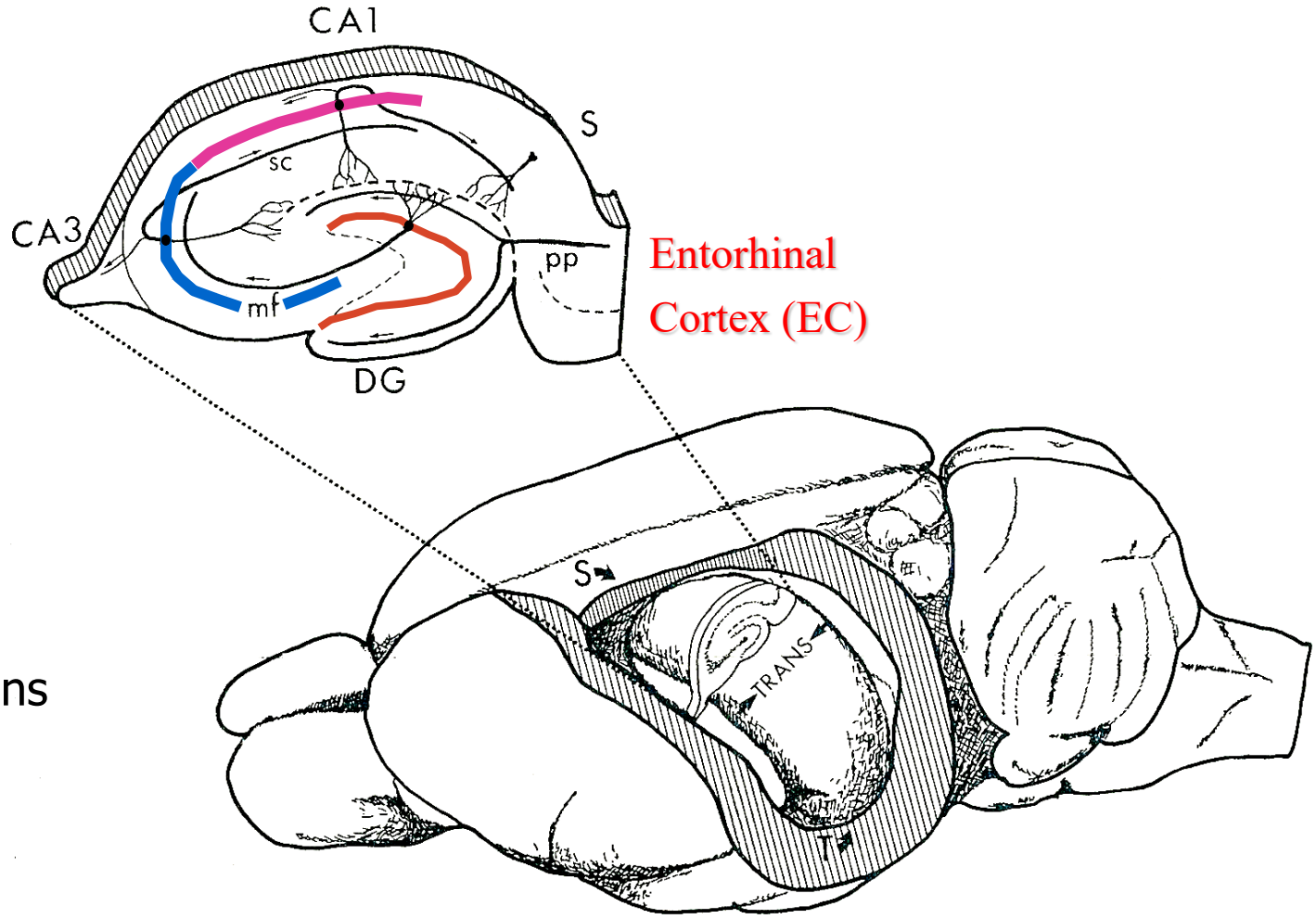
Rat



Echidna
(ancient egg-laying mammal)



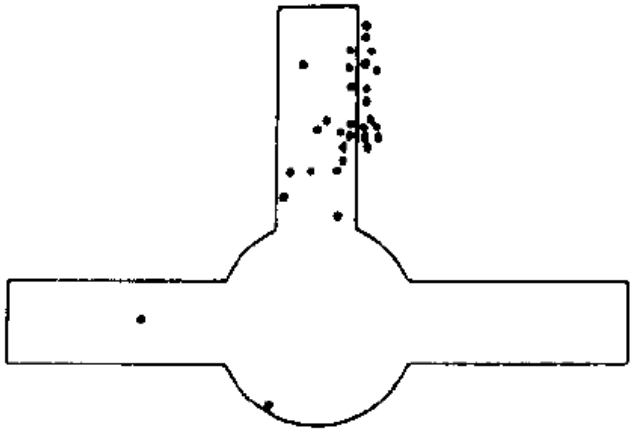
# The hippocampus



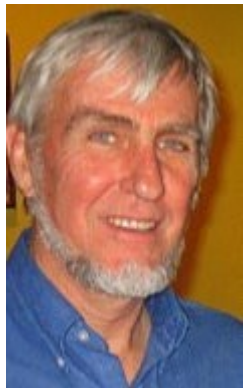

- Highly conserved brain structure across all mammals, including humans (exists also in birds, but looks quite different)
- The most important brain region, clinically

(Amaral and Witter 1989)

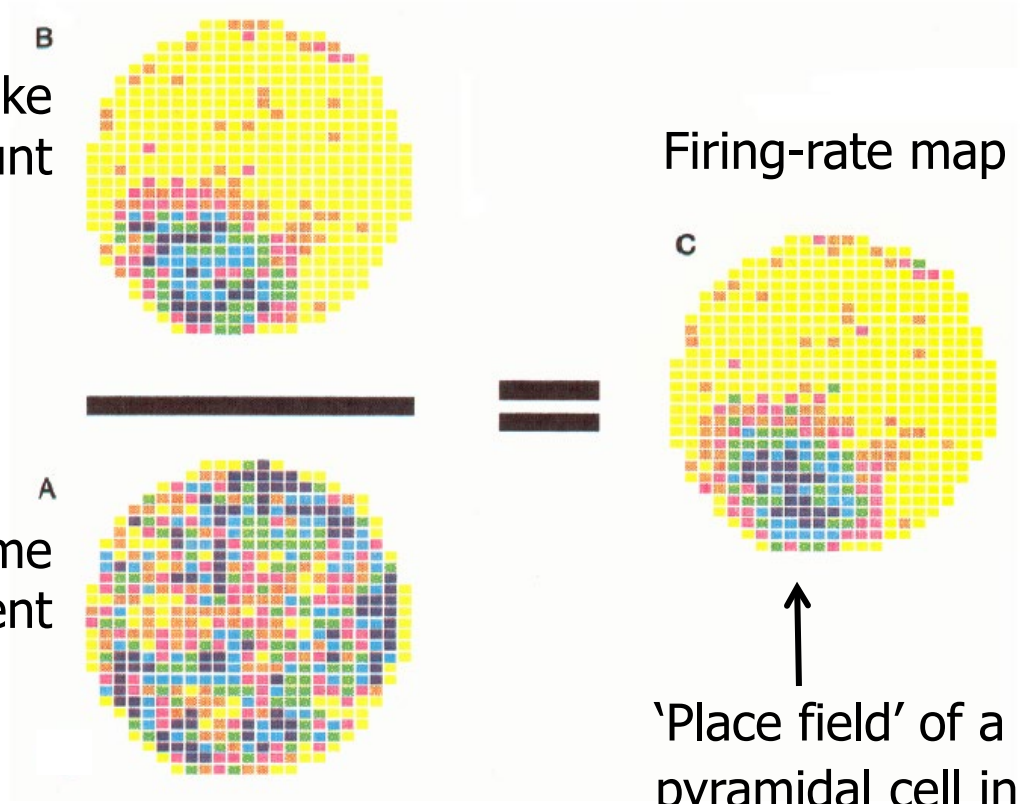
# Hippocampal place cells in rats



(O'Keefe & Nadel 1978)
(O'Keefe & Dostrovsky 1971)



John O'Keefe




'Place field' of a pyramidal cell in rat hippocampus

(Muller et al. 1987)

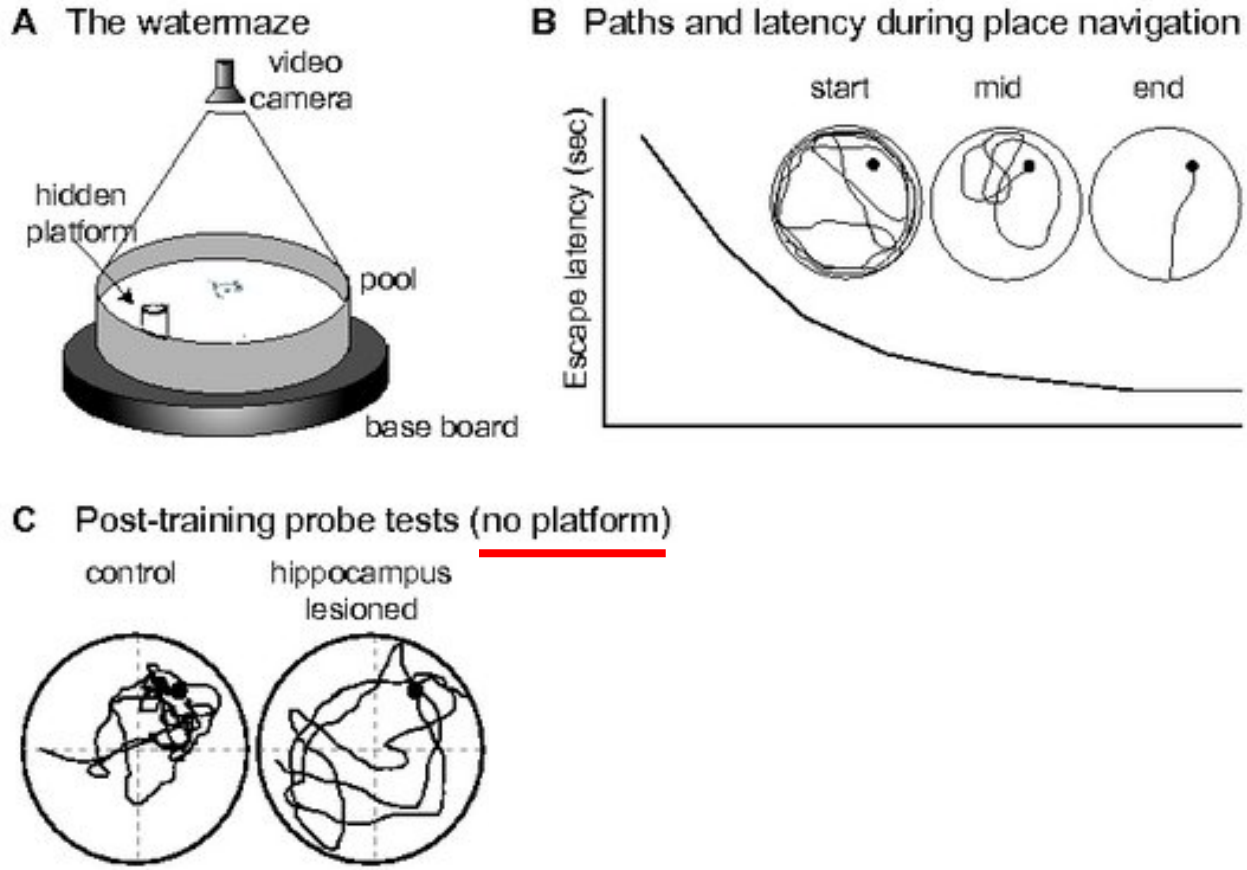
# Movie of a rat hippocampal place cell in action



# Bilateral hippocampal lesions impair allocentric navigation



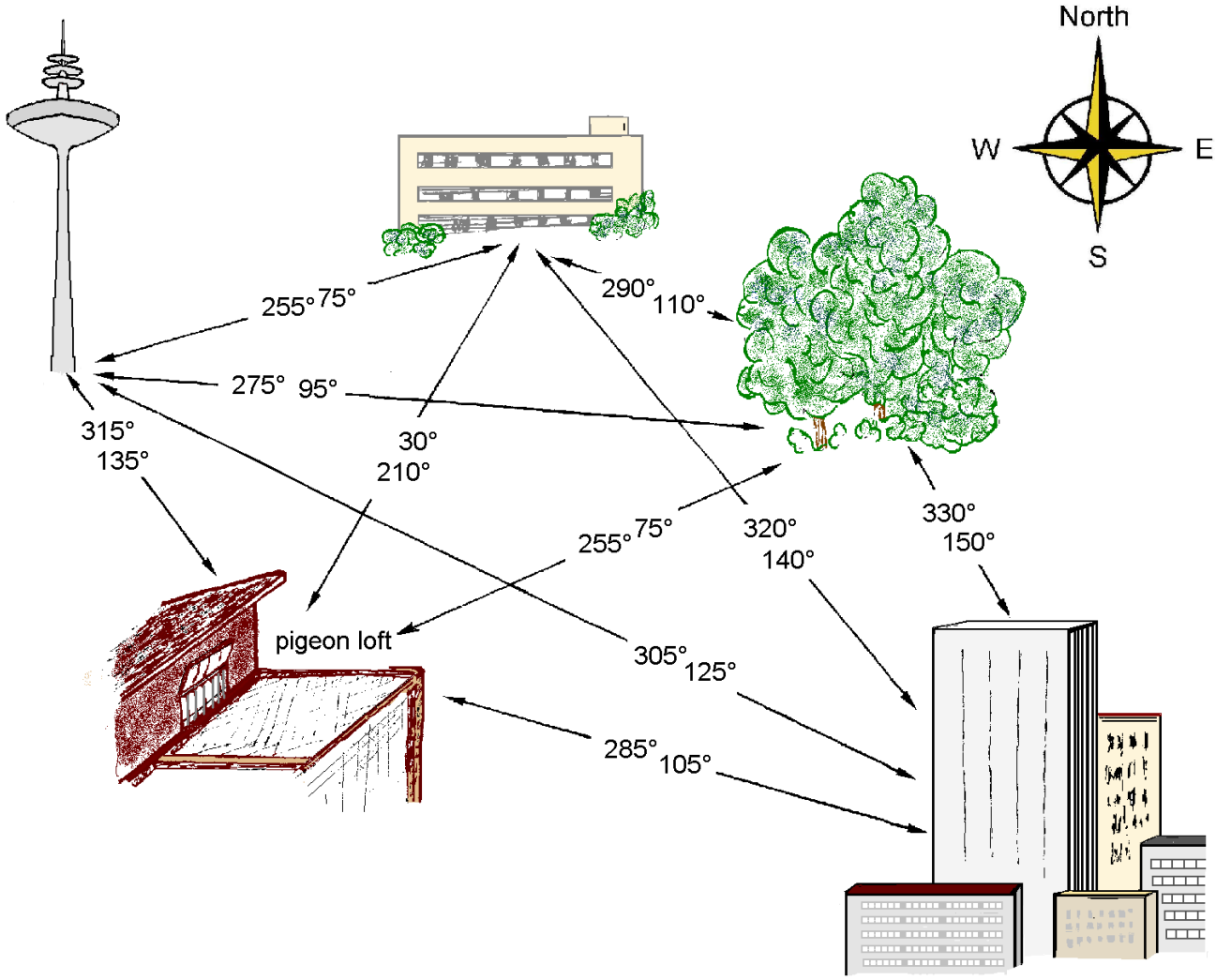
Richard Morris



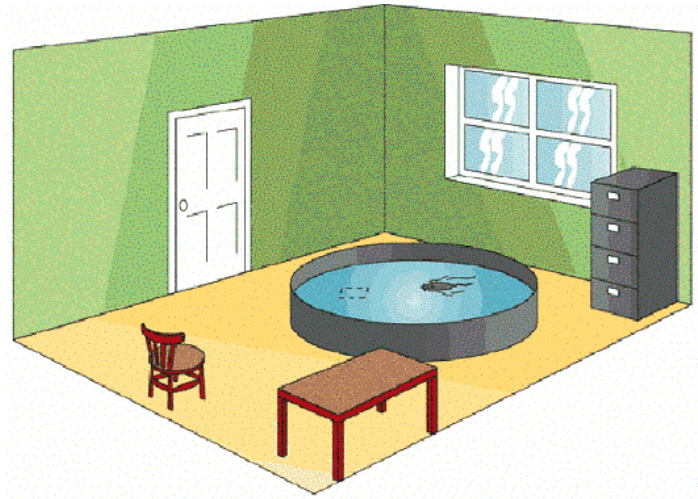

These deficits of spatial memory occur after lesions in dorsal hippocampus – not ventral hippocampus.

# Rat navigation in a watermaze is thought to be similar to the concept of 'Mosaic Map' in birds: self-triangulation based on distal landmarks

Large-scale navigation by birds

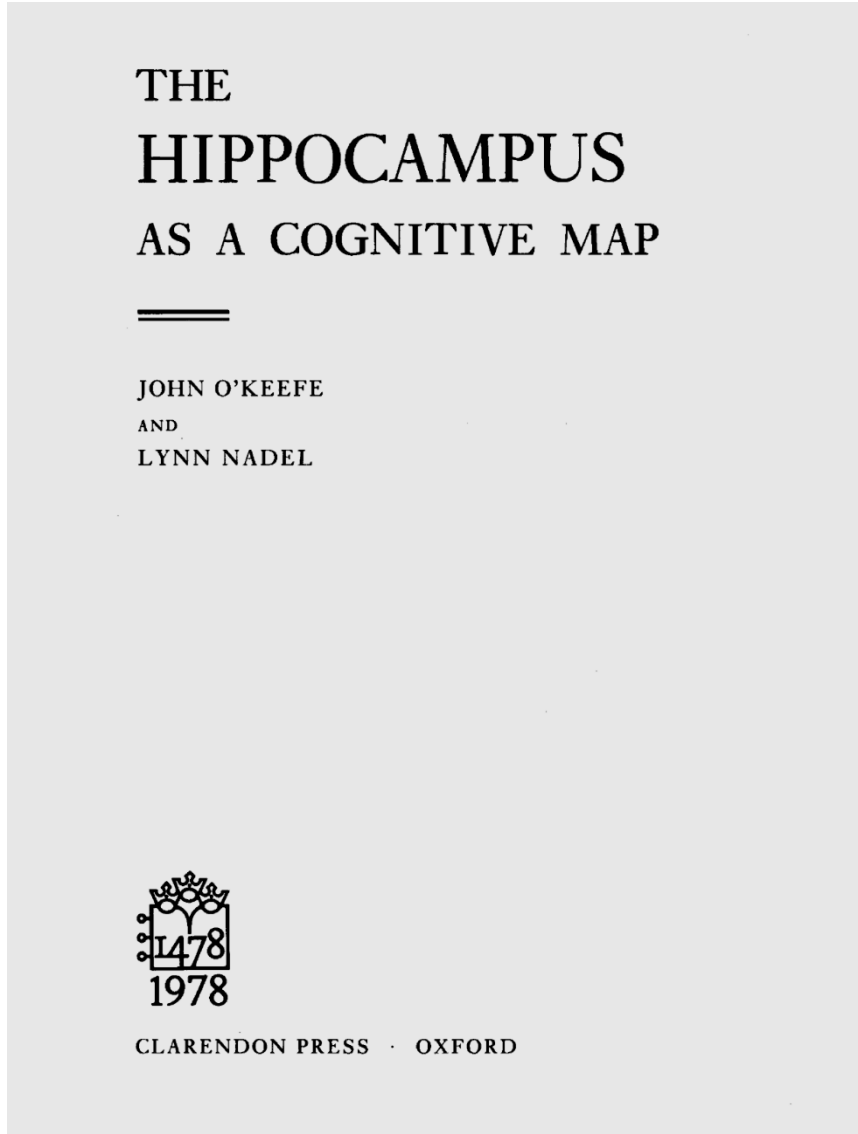



Small-scale navigation by rats in a watermaze



# Neuroethology and the discovery of place cells

## O'Keefe & Nadel, "The hippocampus as a cognitive map" (1978)



THE
HIPPOCAMPUS
AS A COGNITIVE MAP

JOHN O'KEEFE
AND
LYNN NADEL

1478
1978

CLARENDON PRESS · OXFORD

# Neuroethology and the discovery of place cells

## O'Keefe & Nadel, "The hippocampus as a cognitive map" (1978)

question, even with a good bit of luck and insight.* We suggest that during the exploratory phases of research into the function of a structure it is necessary to use a more information-rich methodology, the neuroethological one.

The neuroethological approach differs from the neuropsychological one in several respects. First, it seeks to study the activity of single units in as naturalistic a setting as possible, in the belief that an animal's behaviour in its natural environment maximizes the possibility of producing changes in unit activity that are meaningfully related to that unit's function. It thus embodies the reasonable assumption that the brain of a particular animal is built to operate in a specific environment. At the very

• • •

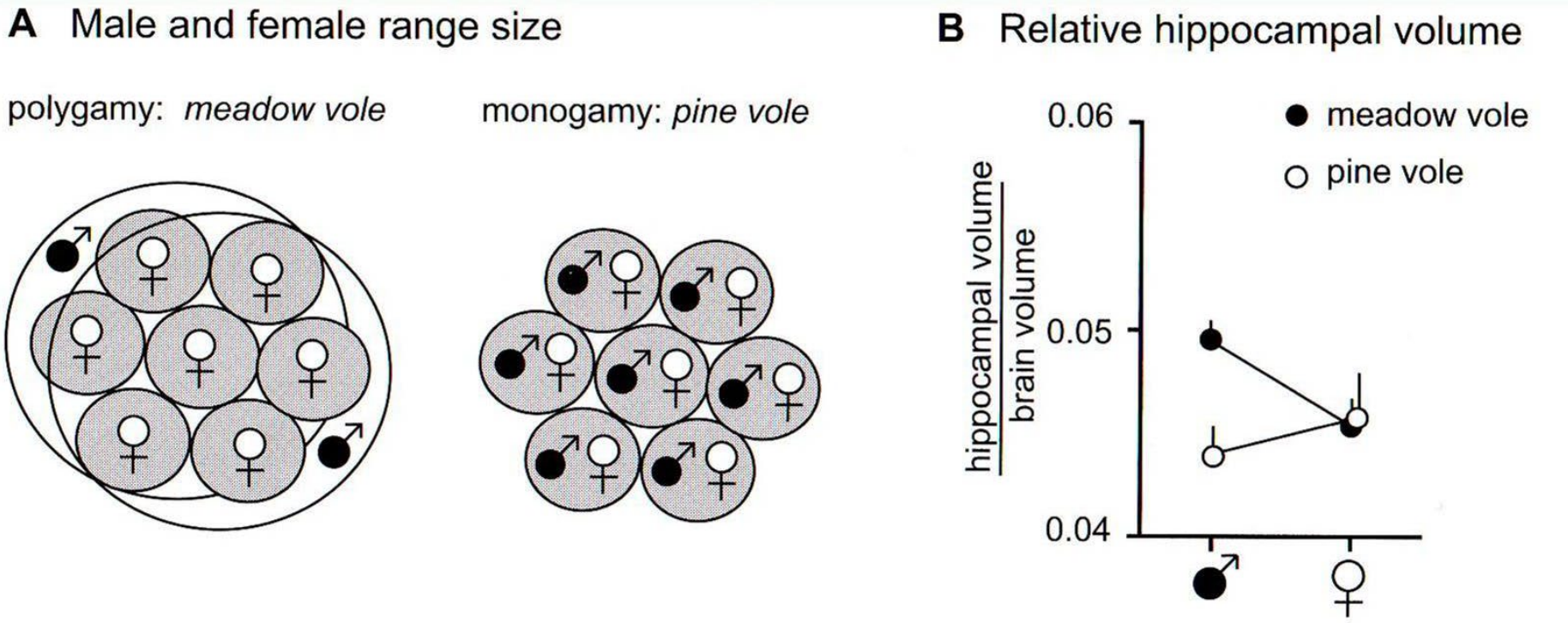
4.7.1. A NEUROETHOLOGICAL STRATEGY

The strategy used in our own work on single-unit activity in the hippocampus of the freely moving rat leans towards the neuroethological, rather than the neuropsychological, approach. The following is a general outline of this procedure.

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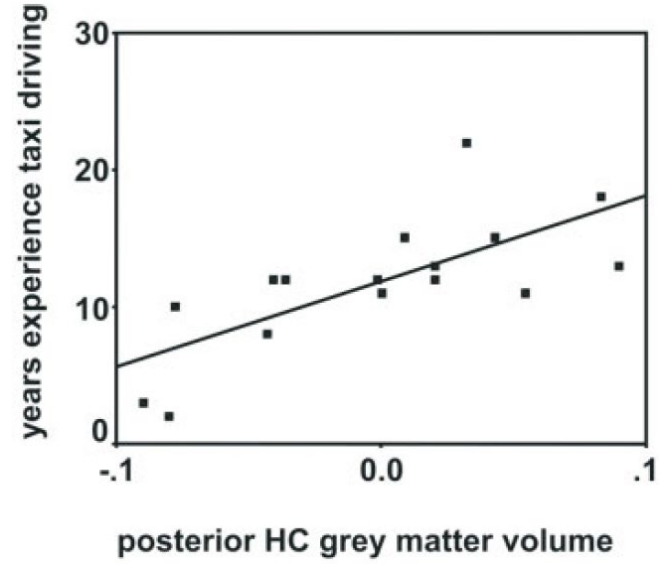
# Hippocampal volume correlates with navigational load in rodents



# Hippocampal volume correlates with navigational load in humans

Volume of posterior hippocampus in humans (equivalent to dorsal hippocampus in rats):

- Larger in London taxi drivers than in age-matched controls.
- Correlated with time spent as a taxi driver.
- Larger in Taxi drivers than in experience-matched Bus drivers.
- In Bus drivers, no correlation with experience was found.




Maguire et al., *PNAS* (2000)

Maguire et al., *Hippocampus* (2006)

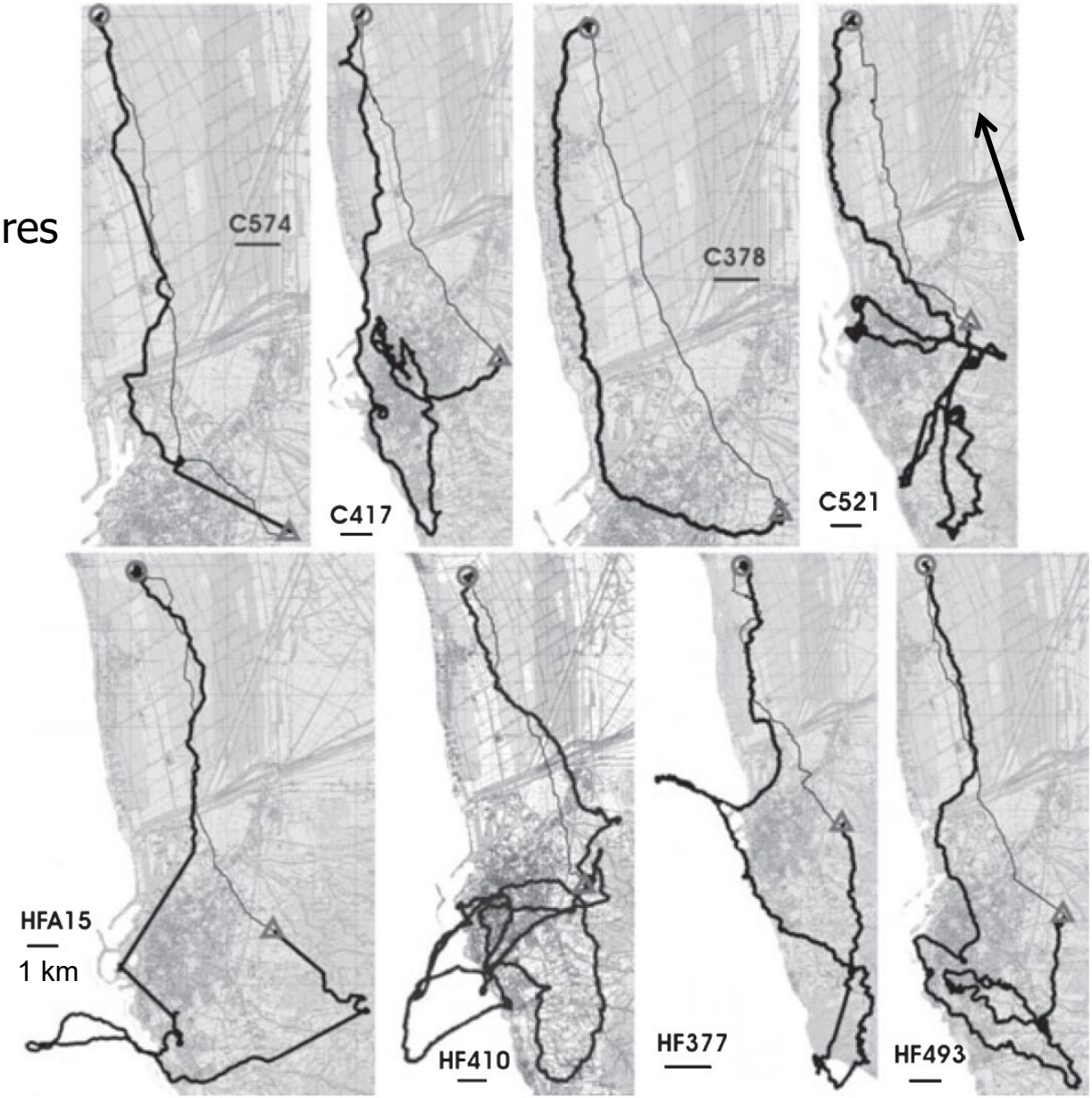
Interpretations:

- The hen and the egg problem: Does posterior hippocampus grow with experience (plasticity), or is a large hippocampus needed in order to do well and "survive" for many years in the demanding profession of a London taxi driver?
- Navigation based on a cognitive-map strategy (taxi drivers) requires/causes a larger hippocampus than route-based navigation (bus drivers) ?

# Lesions in the hippocampus of homing pigeons affect navigation

— Regular release

— Clock-shifted (requires re-orientation)




4 Controls

4 hippocampus lesioned birds

Note flight over the sea →

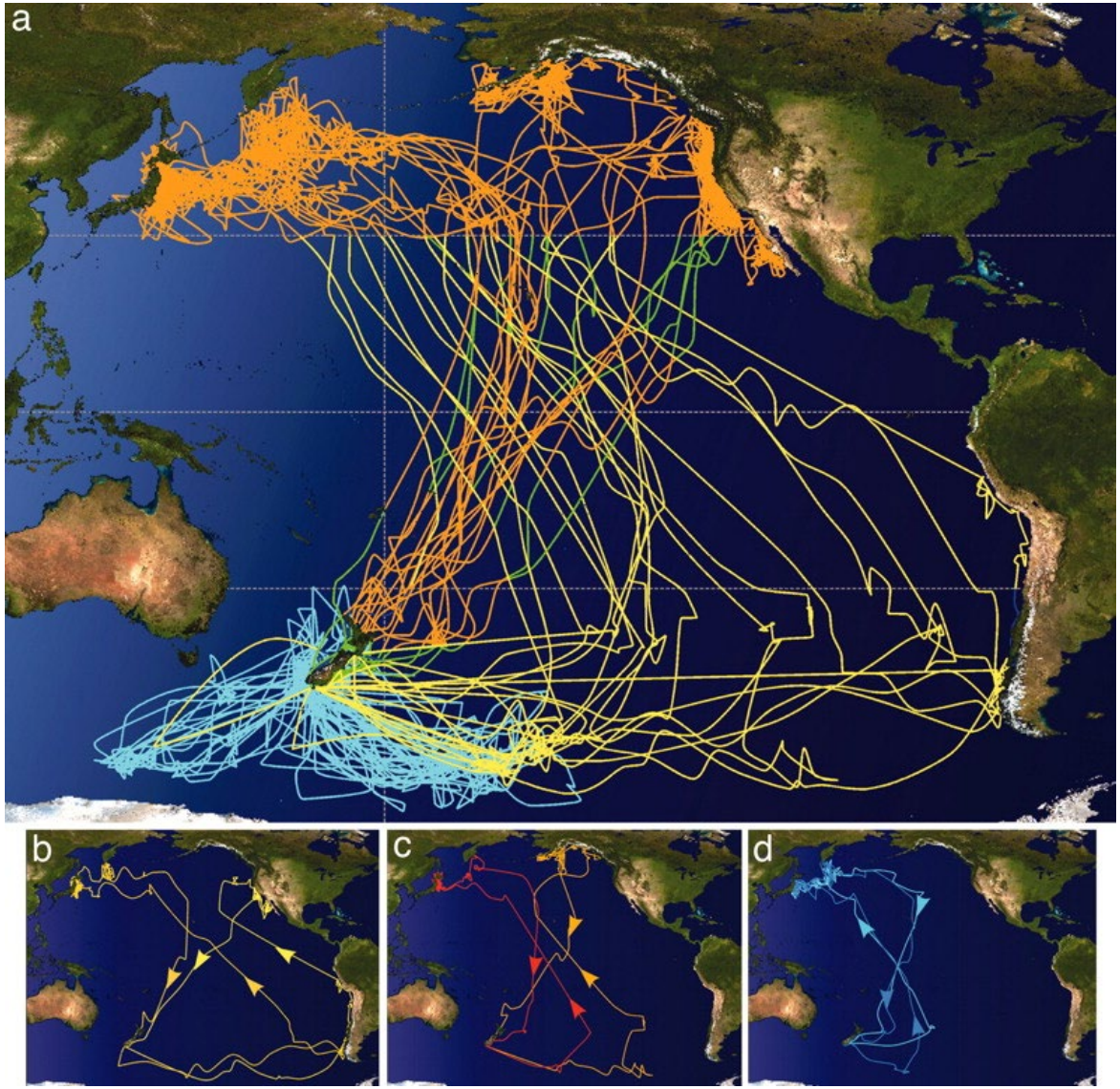
# Lesions in the hippocampus of homing pigeons affect navigation

- *Interpretation*: The map is not stored in the hippocampus, since hippocampus-lesioned birds could home; only re-orientation seems to depend on the hippocampus
- *Caveat*: Bird hippocampus differs substantially in morphology from mammalian hippocampus

Extraordinary flight of a hippocampus-lesioned bird above the sea: Never occurs in normal birds



# *CAVEAT*: No studies of place cells were done on this scale…






Shaffer et al. *PNAS* (2006)

# … and not even on this scale



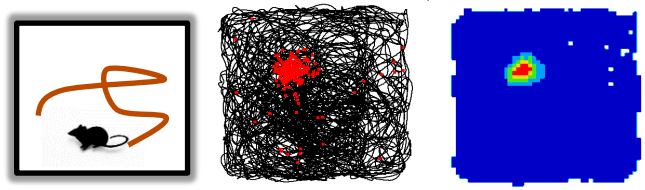
# Brain Mechanisms of Navigation – What I will talk about

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# Neural basis of map-and-compass navigation ?

1. Map → Place cells

Hippocampus

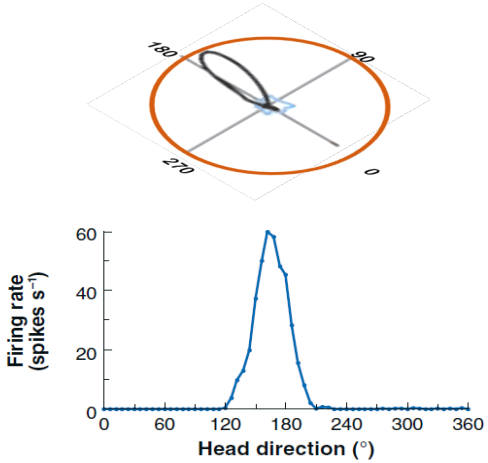


Movie courtesy of Dori Derdikman, 2010

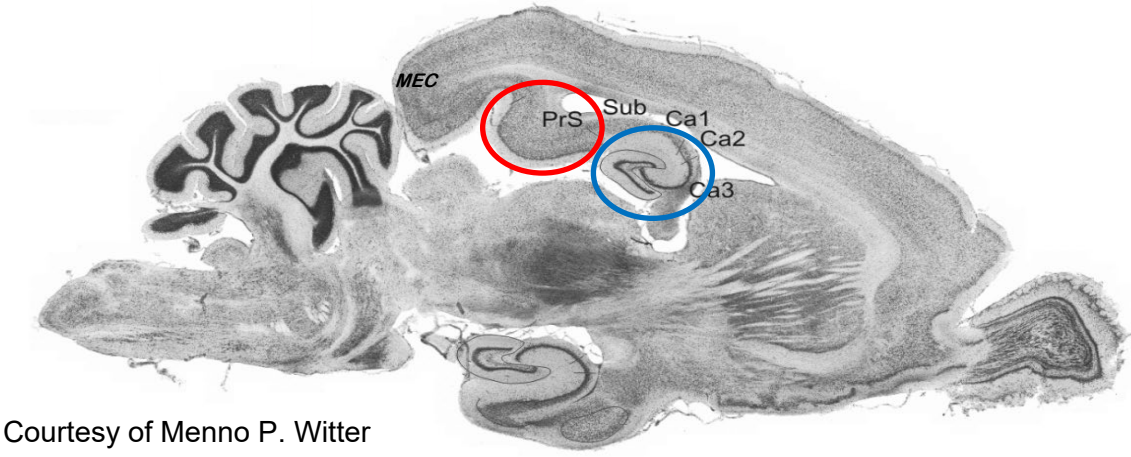
2. Compass

Head-direction cells

Presubiculum (PrS)




Ranck & Taube
JNS 1990



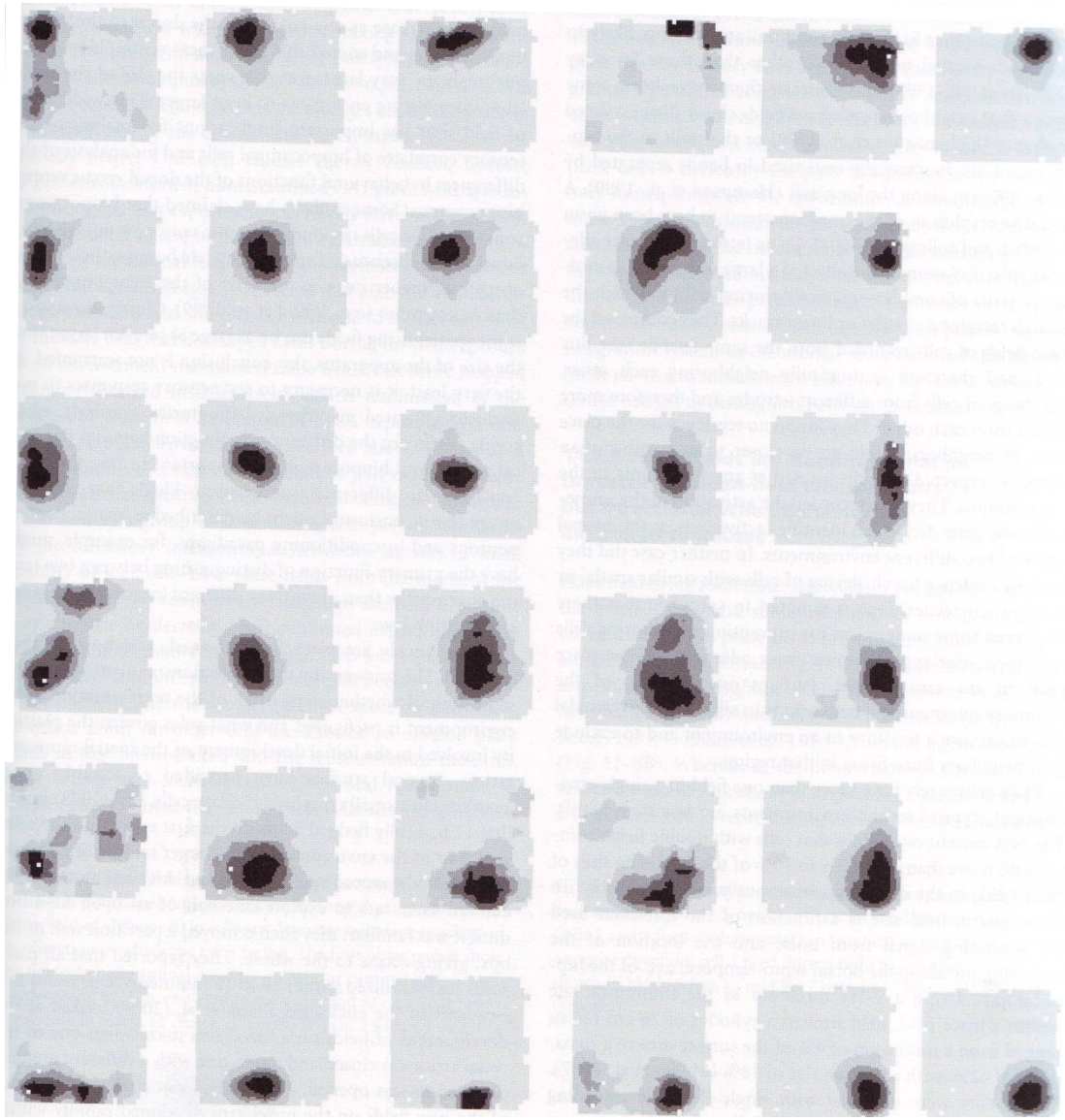

Courtesy of Menno P. Witter

Movie courtesy of Tor Kirkesola, 2010

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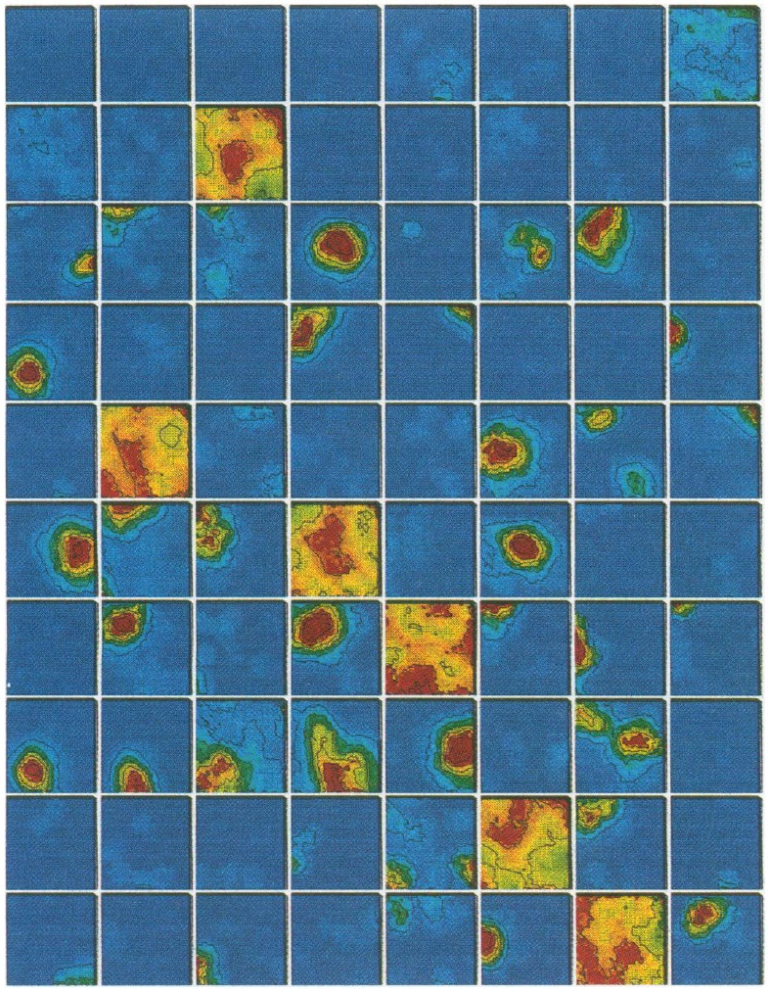
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# The place fields of hippocampal place cells tile the environment

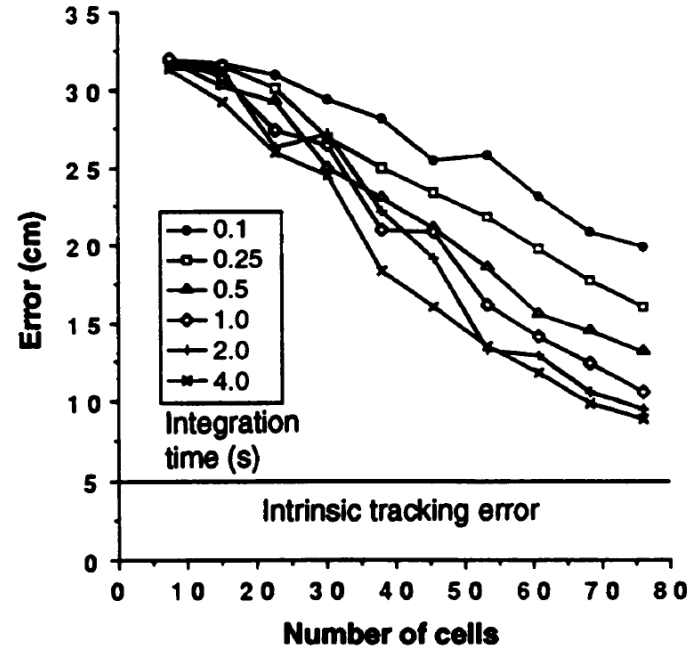


# The rat's location can be reconstructed from the activity of an ensemble of simultaneously-recorded place cells

Tetrode recording of 80 neurons simultaneously





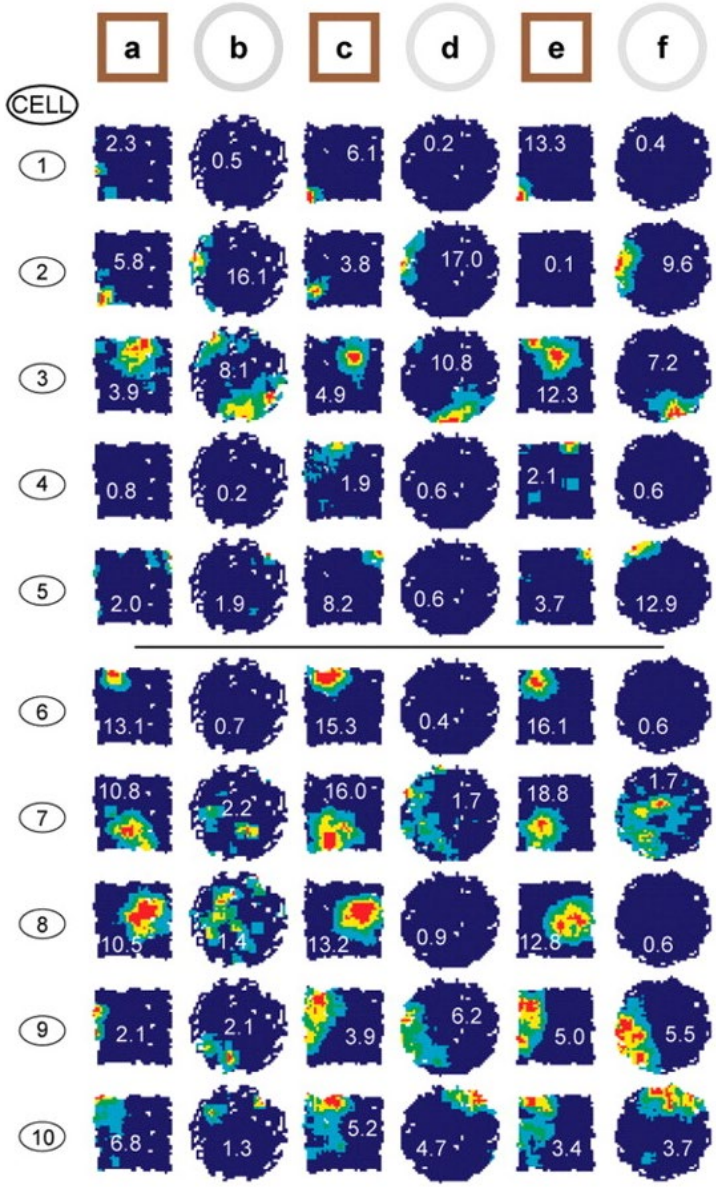



Putative pyramidal neuron (place cell)

Interneuron (very little spatial modulation)

Wilson and McNaughton, *Science* (1993)

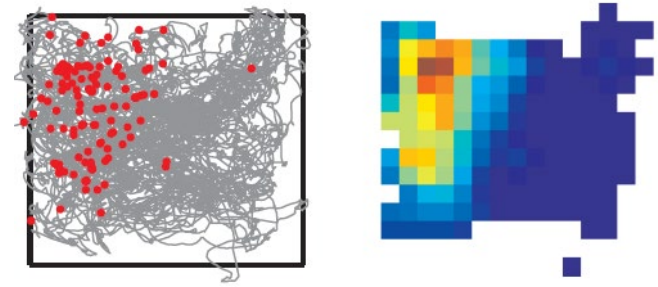
# Multiple maps are stored simultaneously in rat hippocampus



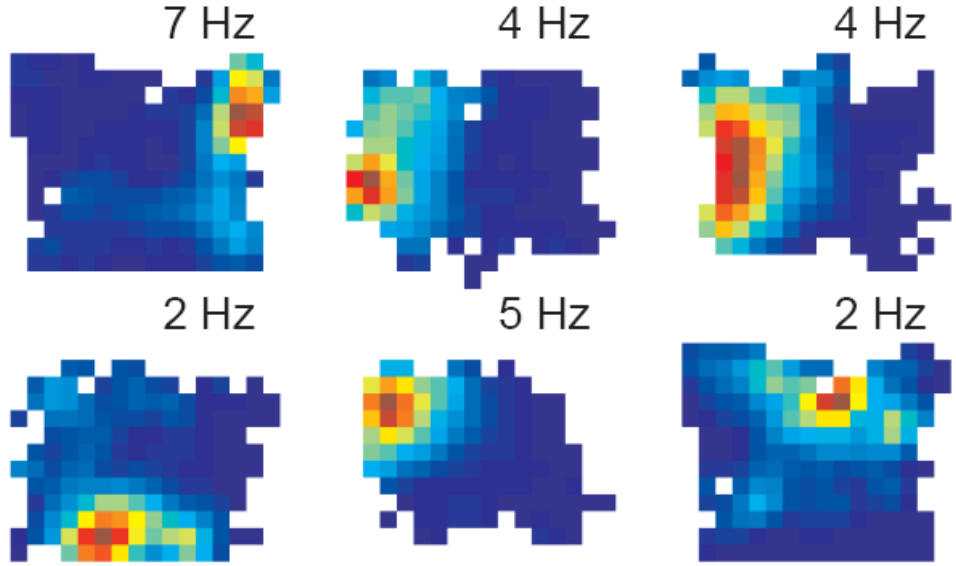

**“Remapping”** between representations of square and circular environments.

Wills et al., *Nature* (2005)

# Place cells in bat hippocampus



A single cell in
big brown bat

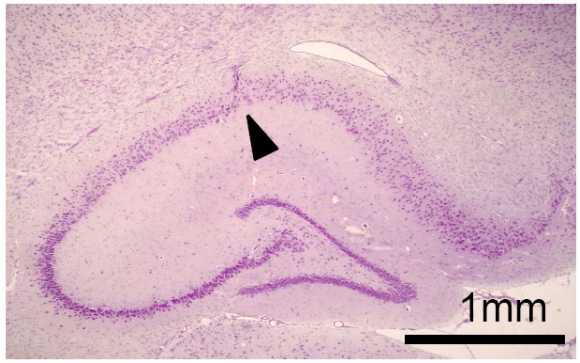


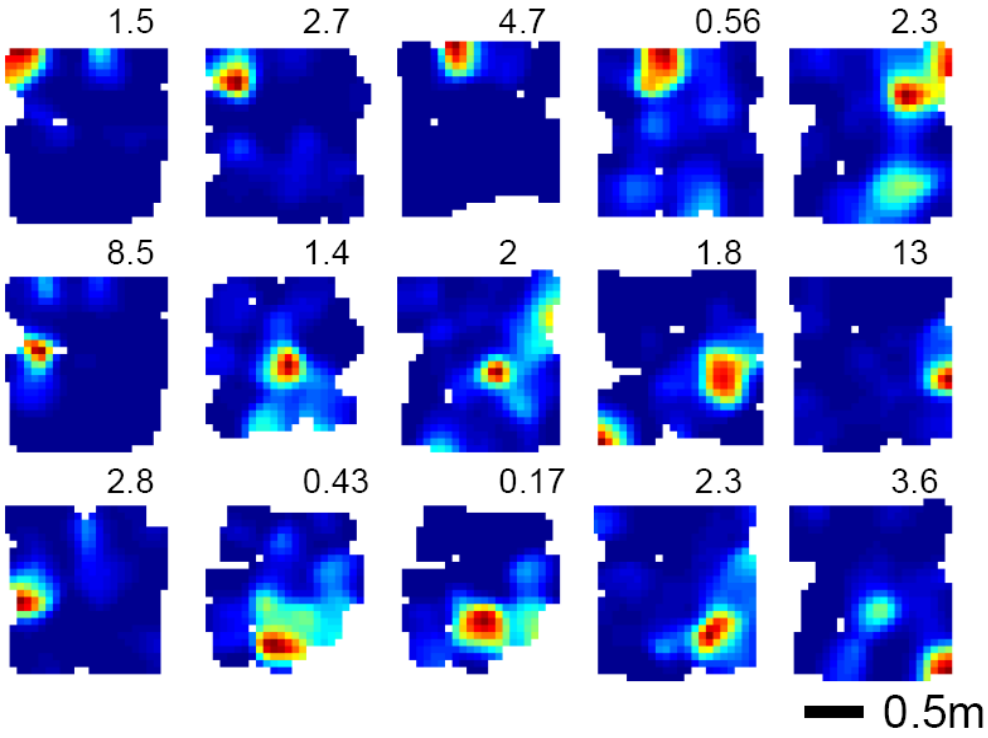

More examples
of place fields
from 6 neurons

Ulanovsky & Moss,
*Nature Neurosci.* (2007)

# And in another bat species: Egyptian fruit bat



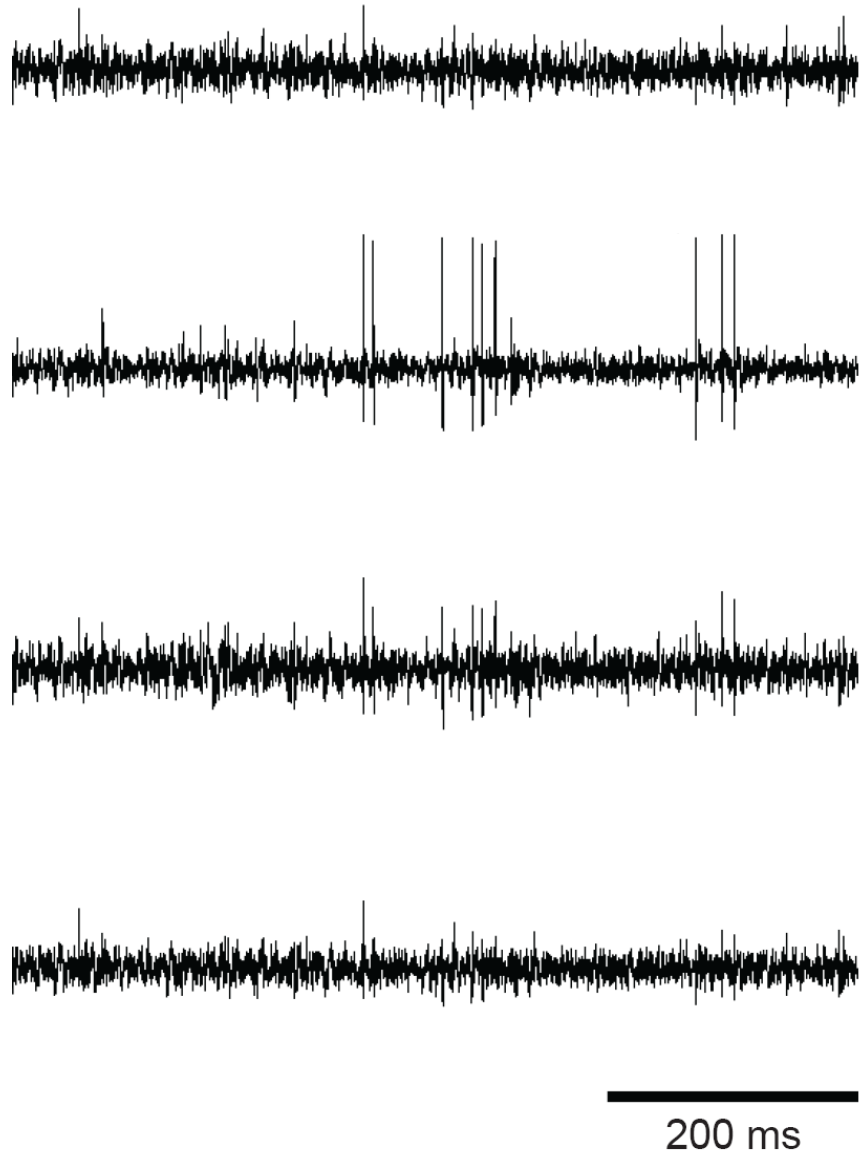



As in rats, these place fields tile the environment, and represent the animal's spatial location.

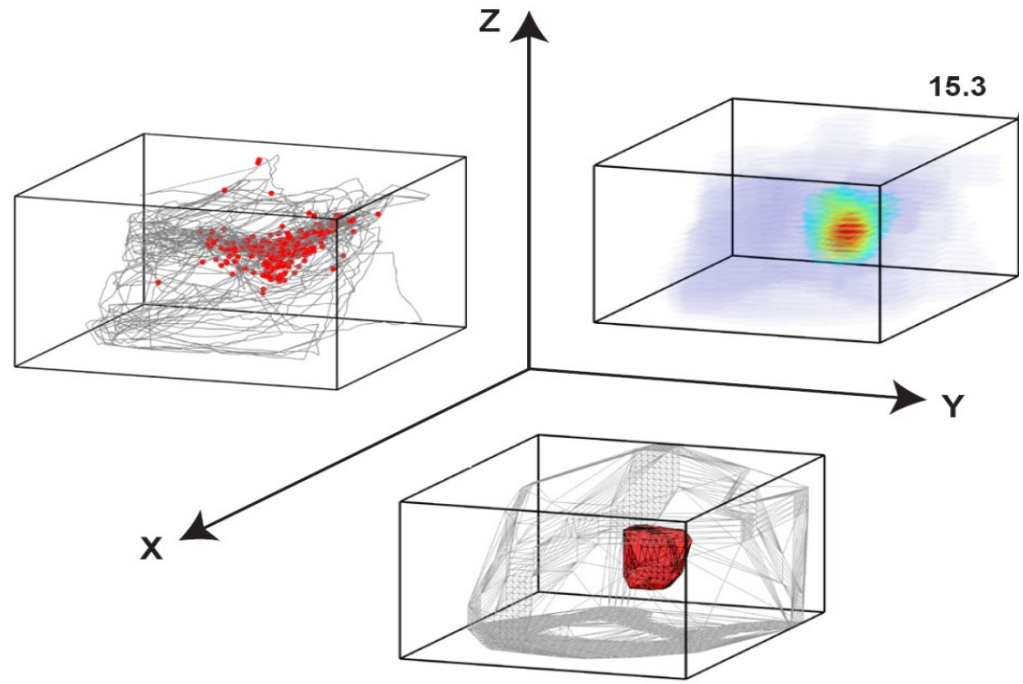
Yartsev, Witter, Ulanovsky
*Nature* (2011)

# Wireless recordings from the hippocampus of a flying bat




Yartsev & Ulanovsky
*Science* (2013)

# 3-D place cells in bats



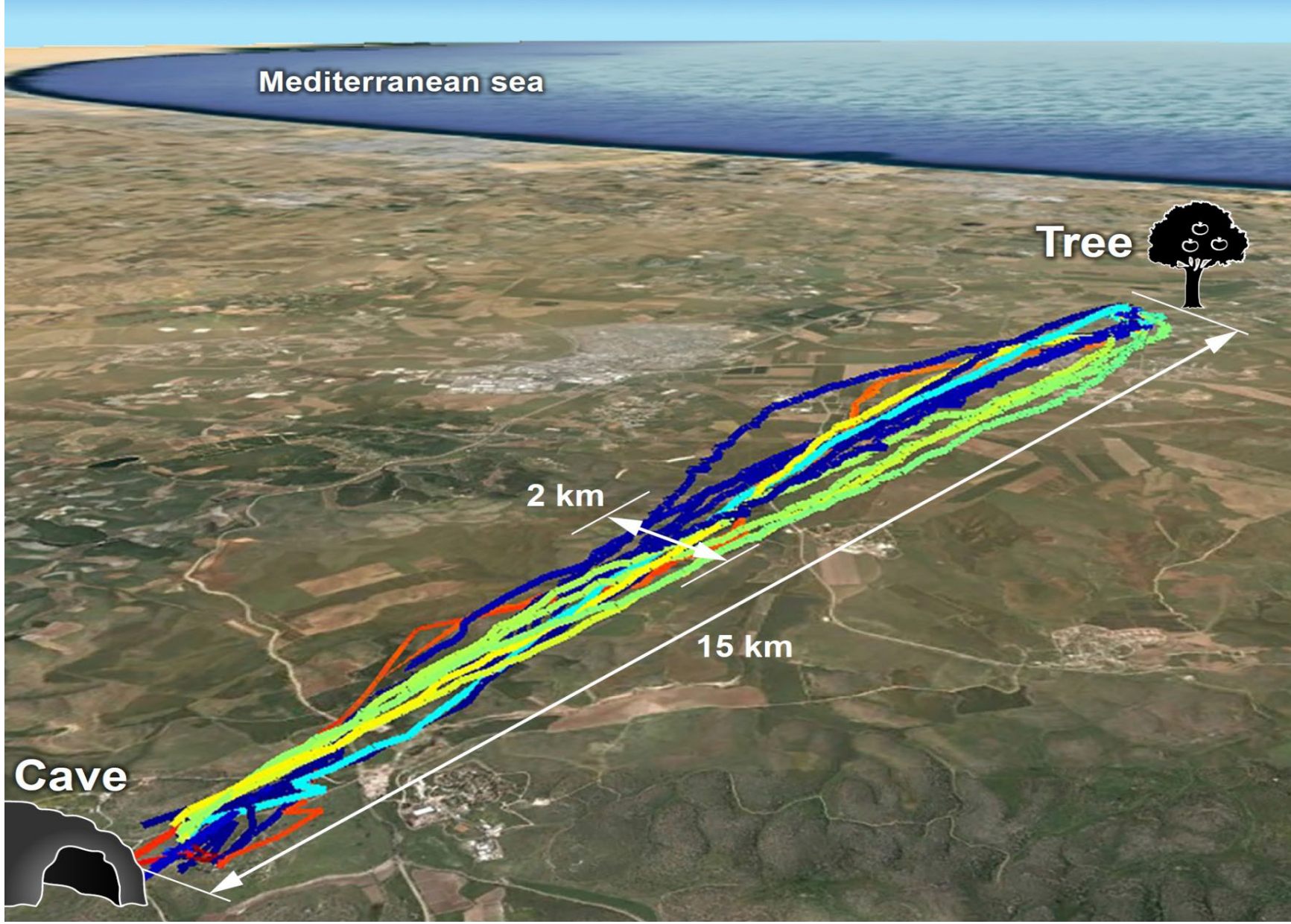

- **3-D place fields are spherical in shape: Same resolution in all directions (<u>isotropic</u>).**
- **3-D place fields Tile space <u>uniformly</u>.**

Yartsev & Ulanovsky
*Science* (2013)

# BUT… Largest straight environment used in rodents: 18 meters (of which ~10 meters within one room)

*Not large enough…* Rats in the wild (real rats, not laboratory rats) navigate distances of up to 1–2 km per night = much larger distances than 10 meters.

**The problem: You cannot tile the natural huge navigational spaces with small place fields. Some other coding must take place.**

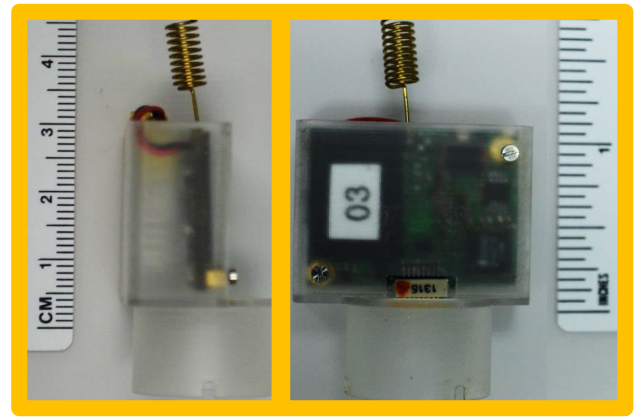


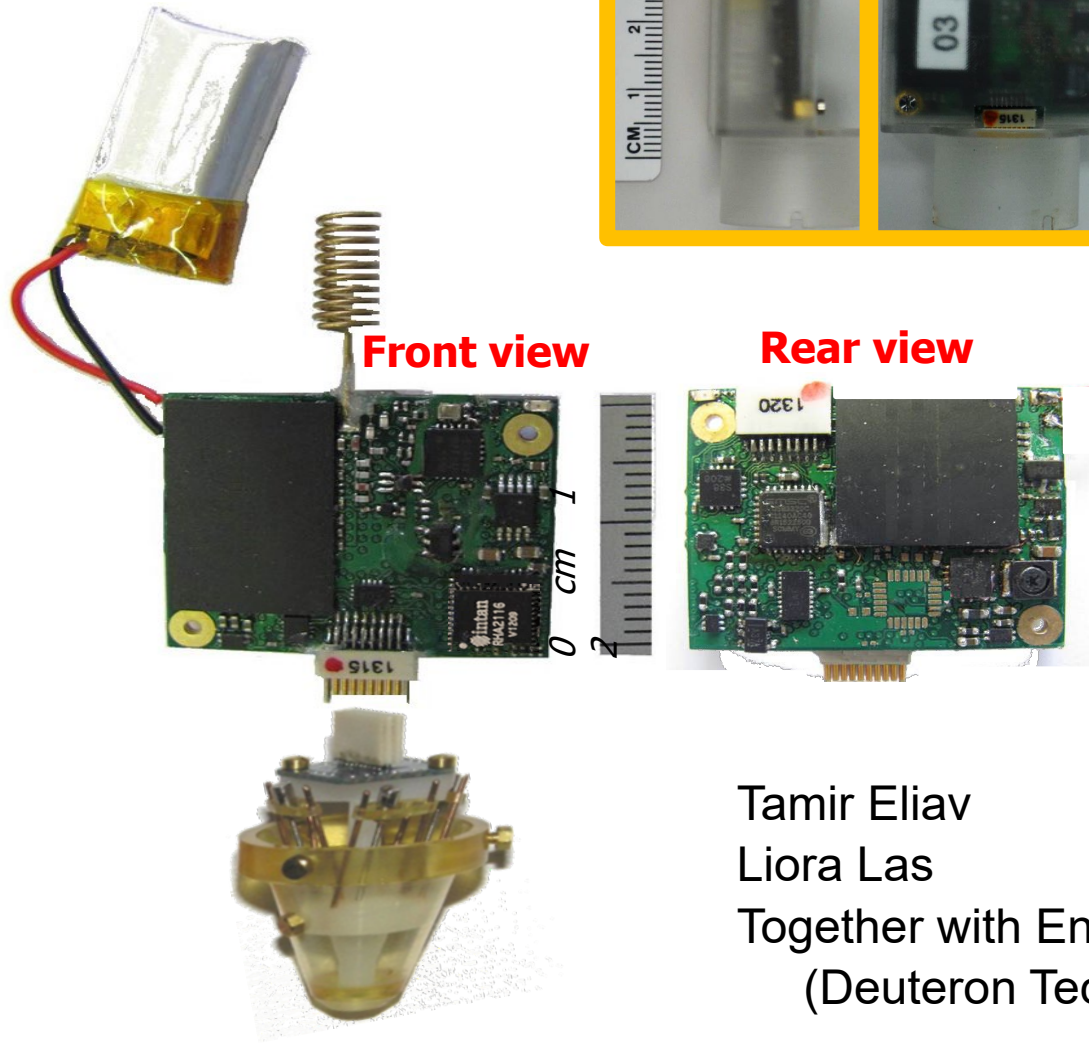

Geva-Sagiv, Las, Yovel, Ulanovsky

- *Nature Rev. Neurosci.*, 2015
- *PNAS*, 2011

# 1. Developing on-board 64-channel neural logger

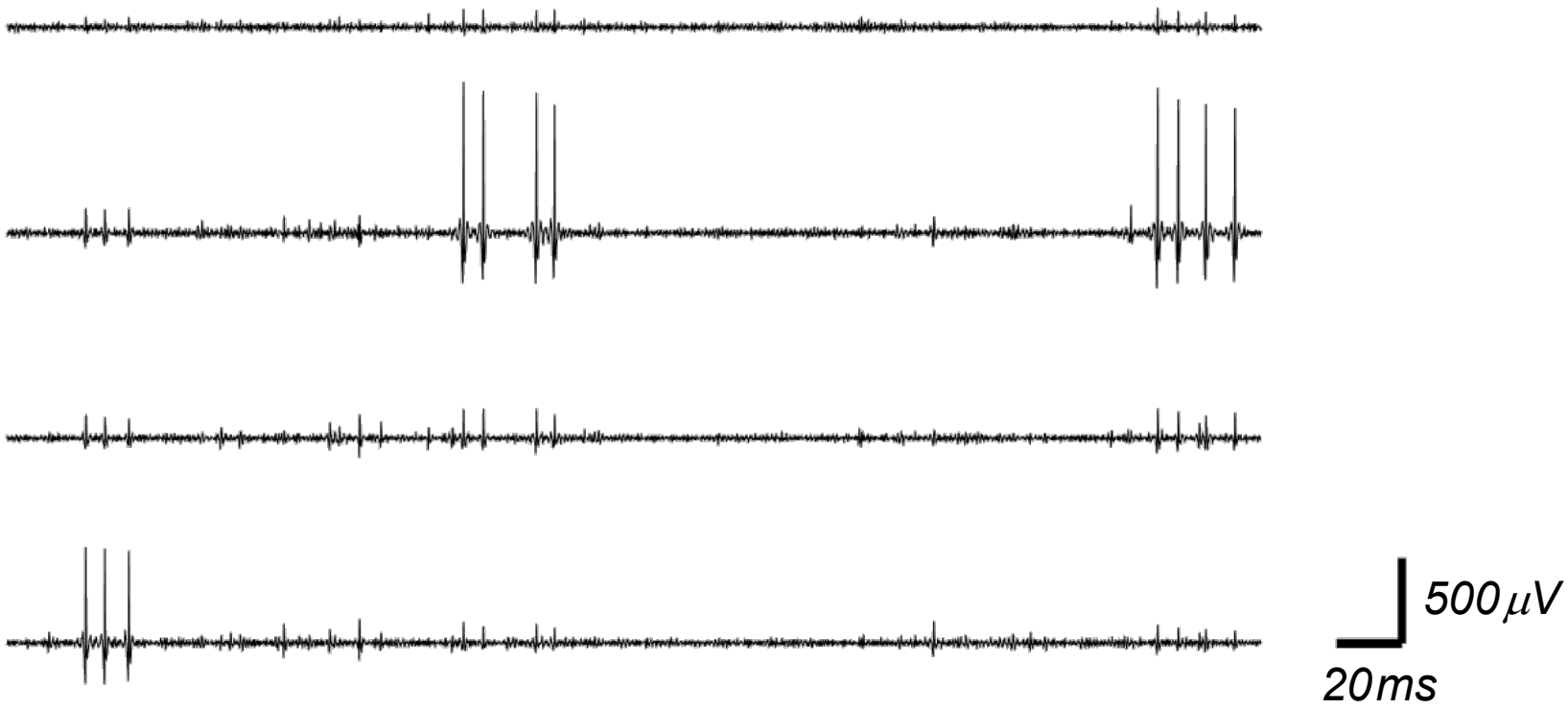





Tamir Eliav
Liora Las
Together with Eng. Jacob Vecht
(Deuteron Technologies)

# 1. Developing on-board 64-channel neural logger



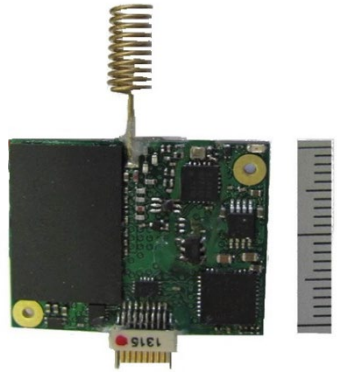

Neural Logger:

- 64 neural channels
- Ultrasonic Microphone
- 9-axis Motion sensor

} An entire on-board Flying ePhys Lab

# 1. Developing on-board 64-channel neural logger

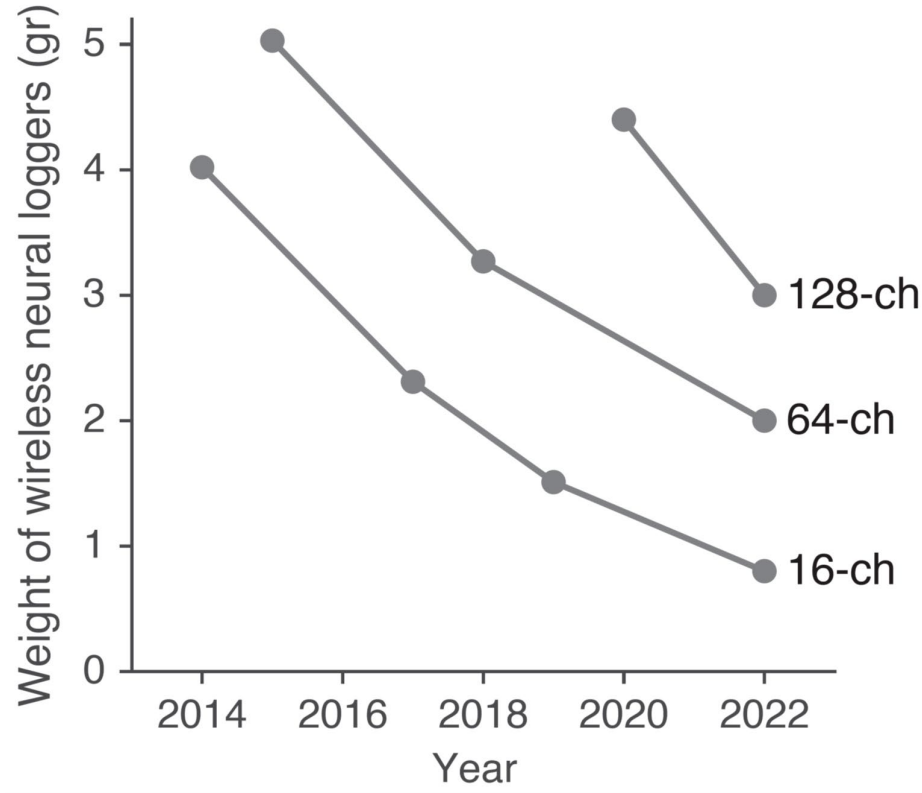
## Technology development



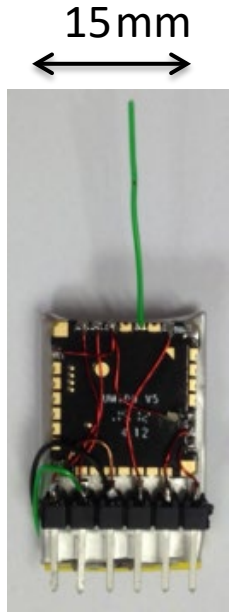
Smallest
wireless neural logger
in the world



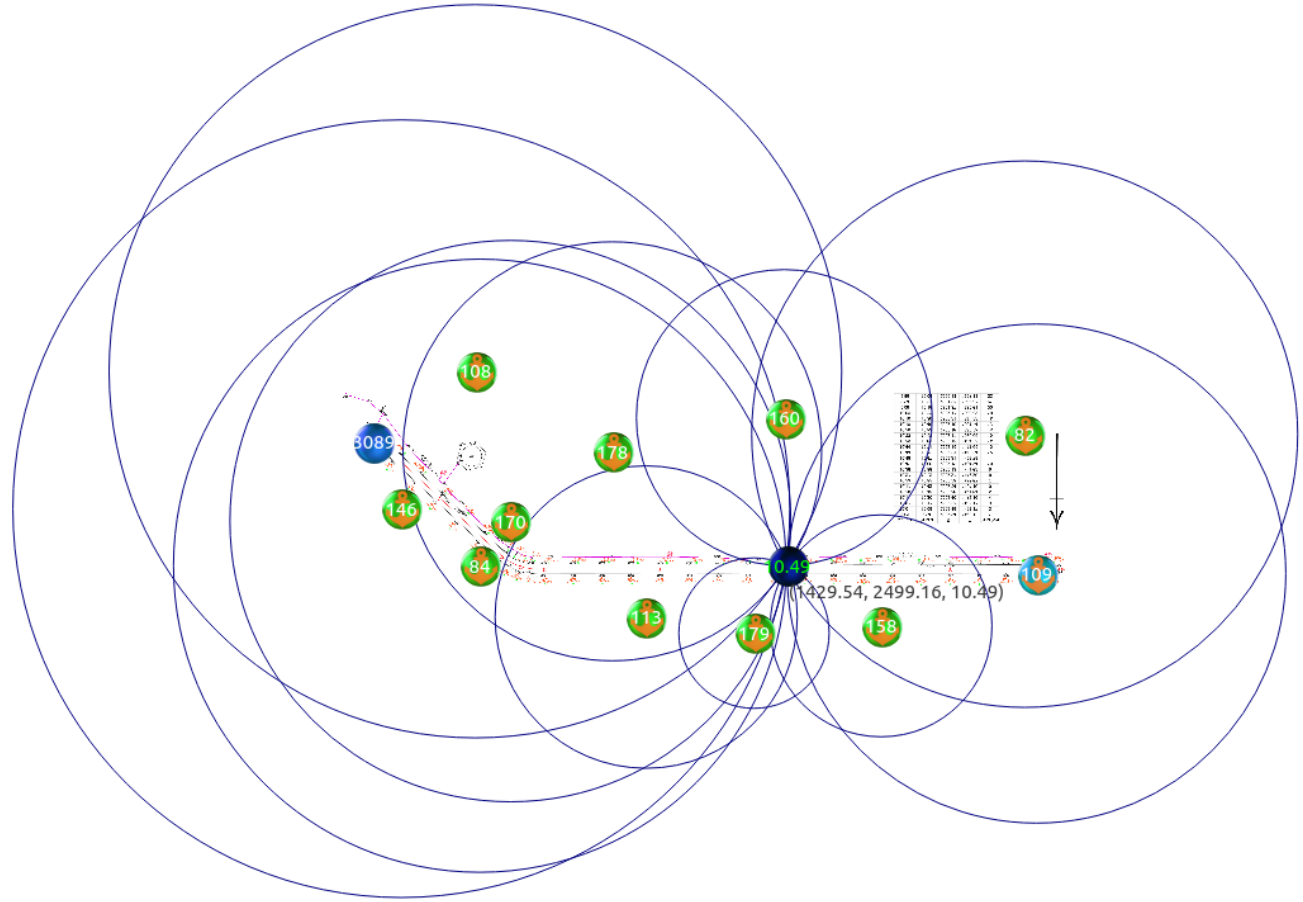
## Rapid miniaturization of wireless technology



# 2. Large-scale localization system with 5-cm precision




BeSpoon, Inc.




0.49 Tag (mobile unit)

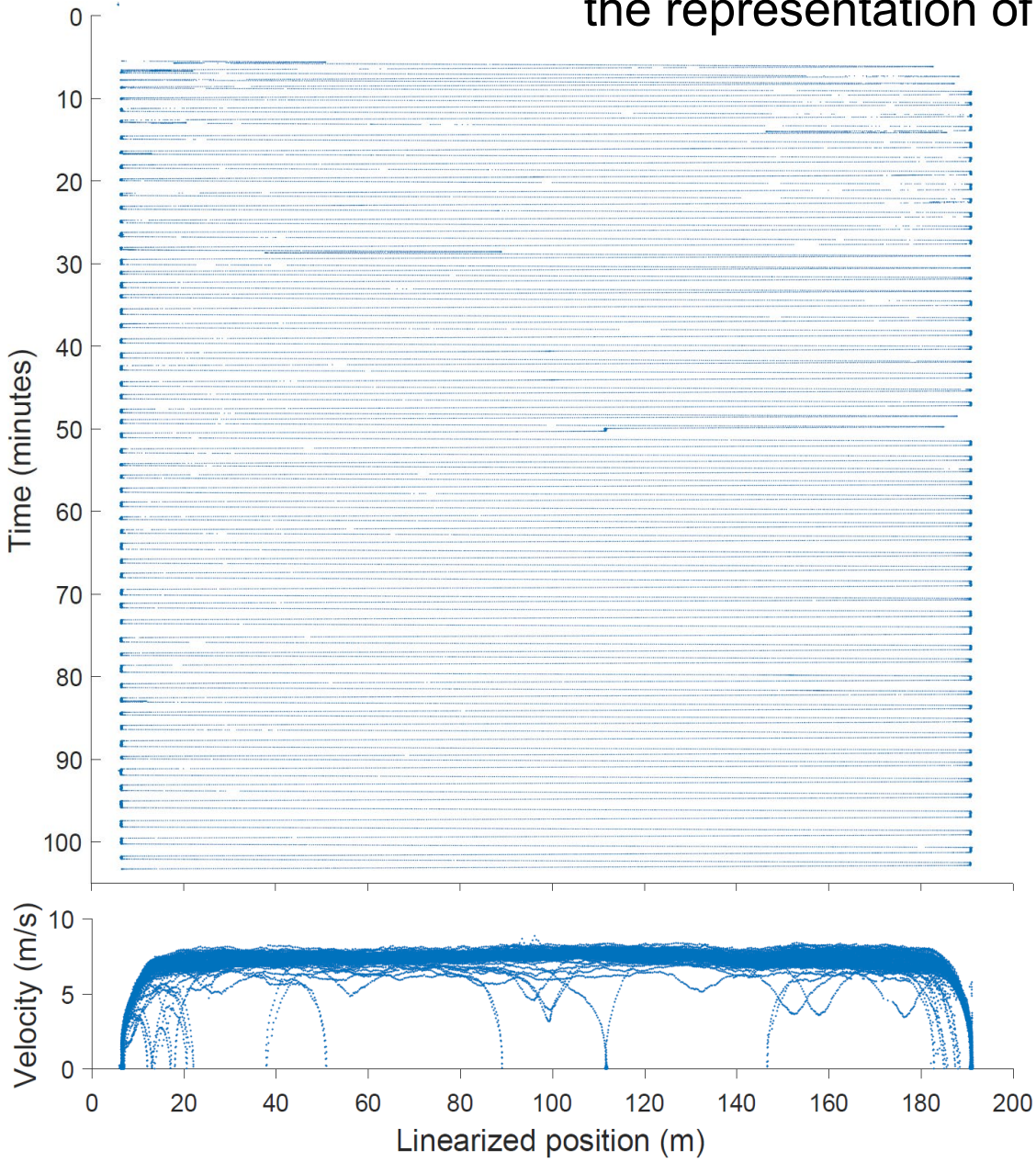
108 Anchor (ground-based)

# 3. Large behavioral setup

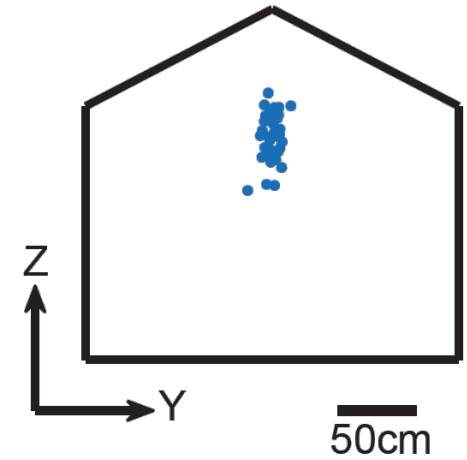


# Behavior

→ Utilizing the bat's flight-speed to measure the representation of large spaces.

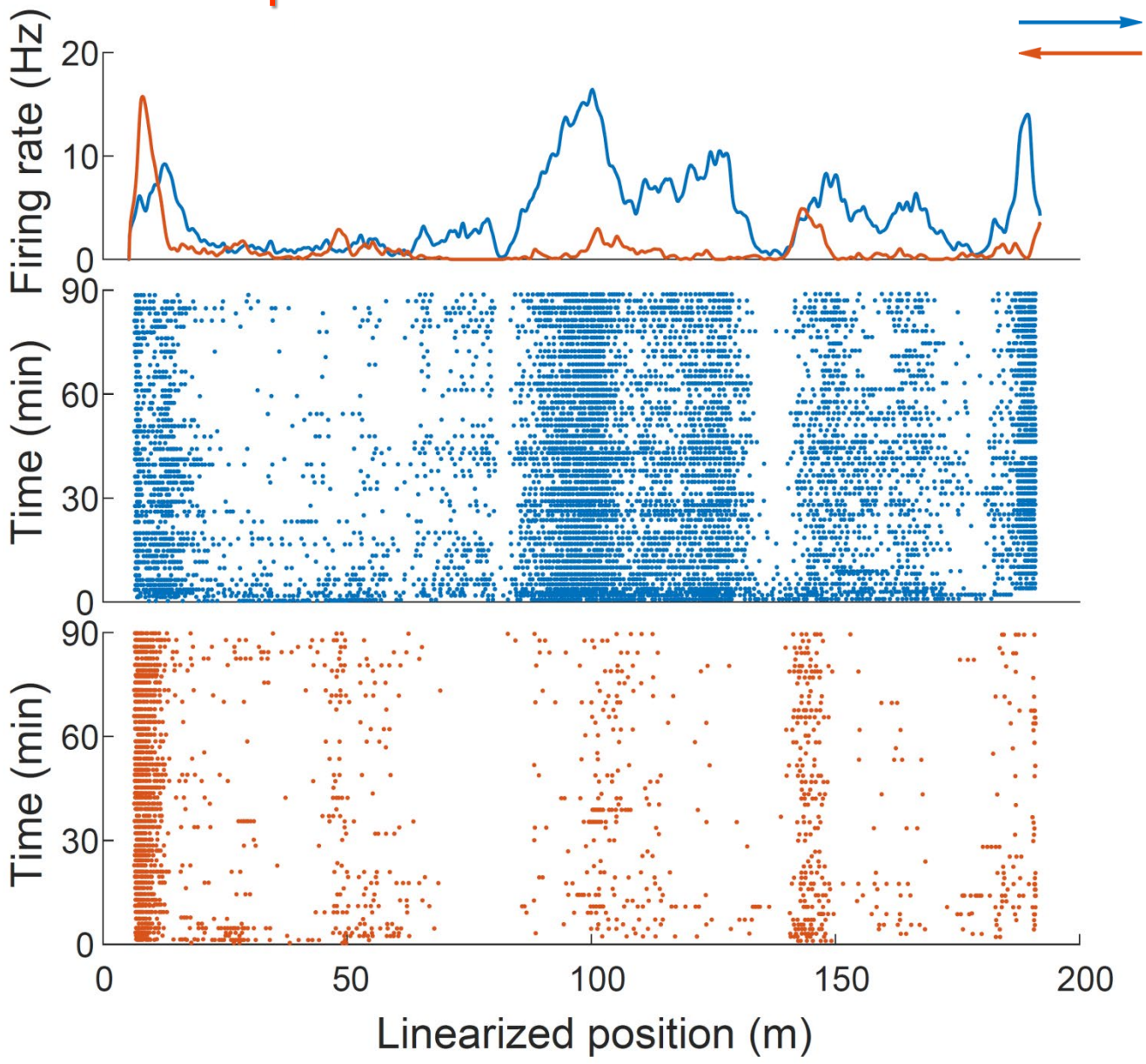



- Food rewards at both ends
- Direct flights from start to end
- 100 laps / session (20 km / 1.5 hrs)



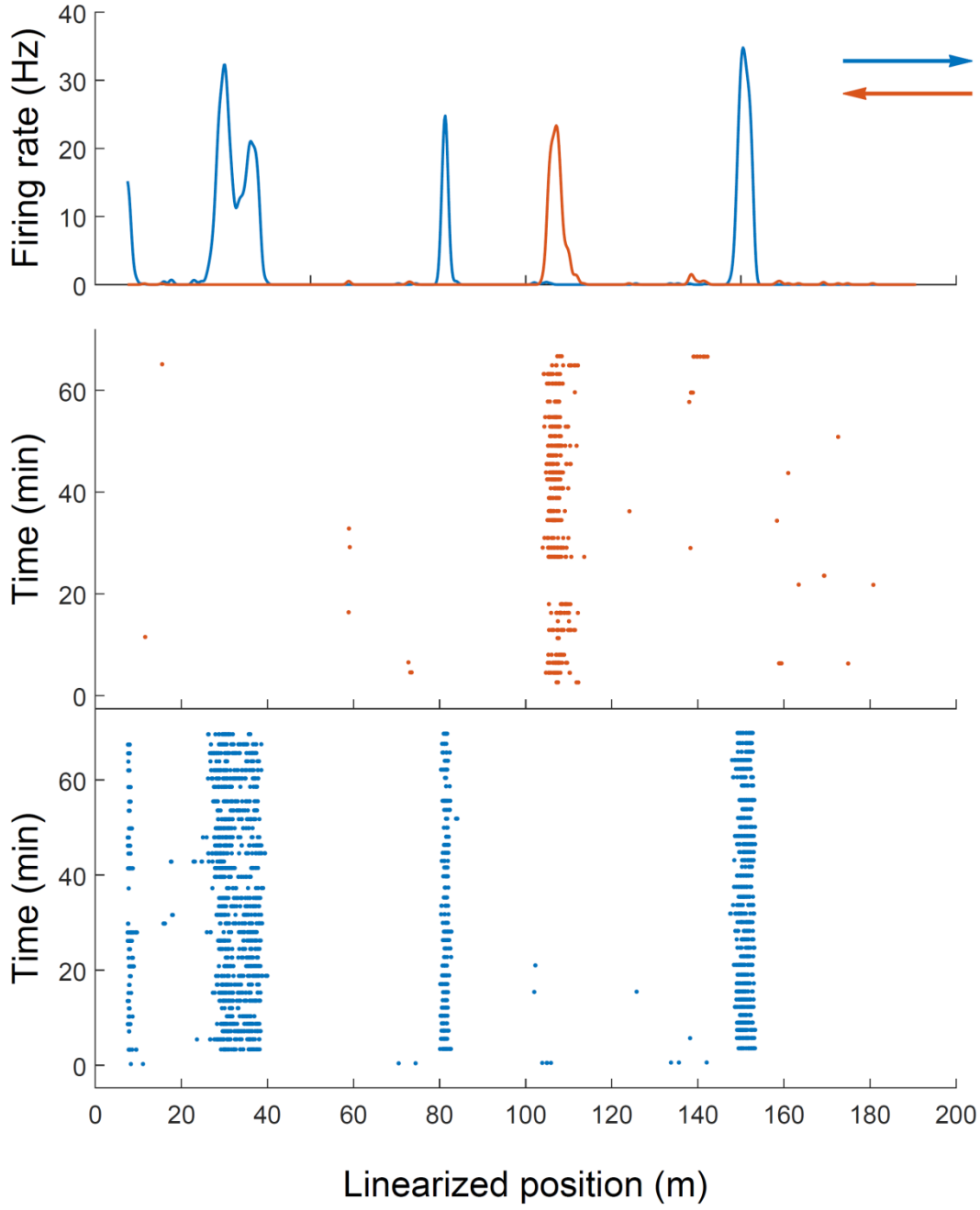

- Flight speeds: 7-8 m/s

# Examples




Example cell from dorsal CA1

Eliav et al. *Science* (2021)




[Example 2]

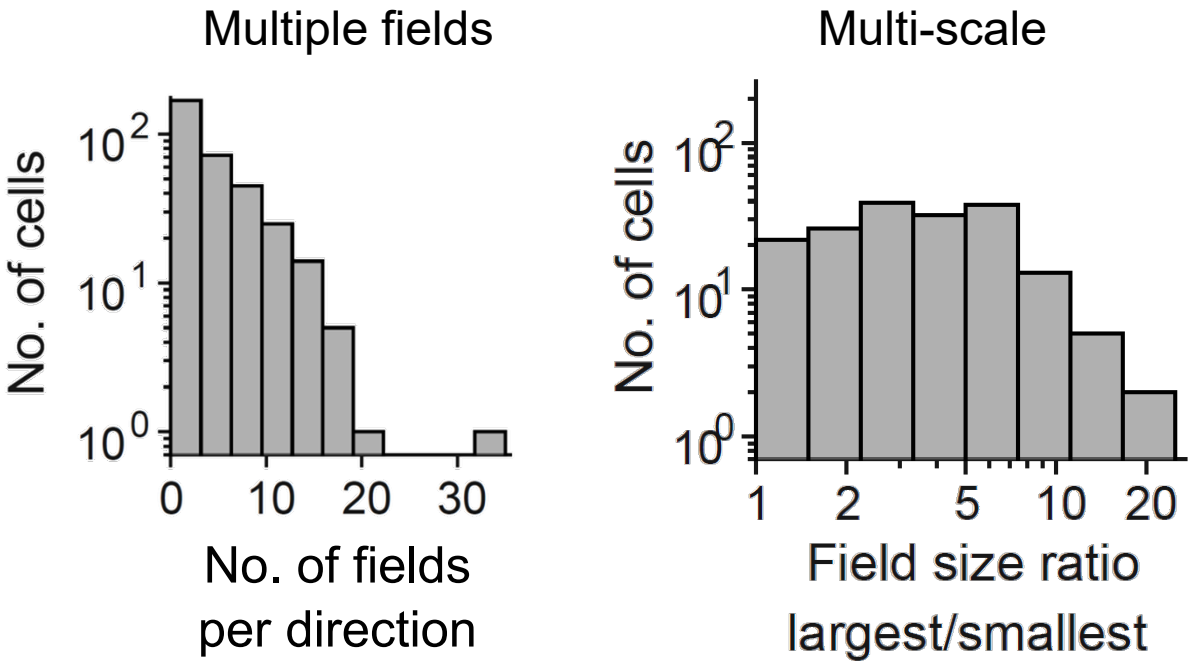
Properties of this spatial code:

- Large fields
- Multiple fields
- Multi-scale

→ Looks very different from place cells recorded in small lab setups !

Eliav et al. *Science* (2021)

# Population




Properties of this spatial code:

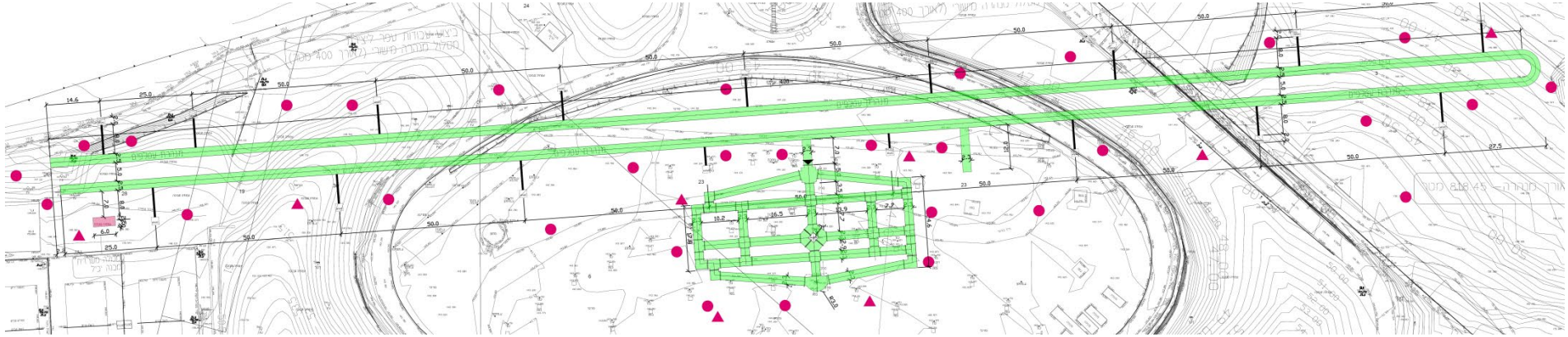
- Large fields
- Multiple fields
- Multi-scale

→ Looks very different from place cells recorded in small lab setups !

- **Theoretical analysis of decoding errors explained <u>why</u> multi-scale coding is better for large environments: Because it allows *much* better spatial coding.**

Eliav*, Maimon*, et al. *Science* (2021)

# Building a 700-meter tunnel + large Maze (60 x 35 meters)



- **Tunnel:**
  - Coding of large spaces across hippocampal subregions.
  - Effects of history.

→ Recording large ensembles of neurons in-flight.

# Building a 700-meter tunnel + large Maze (60 x 35 meters)



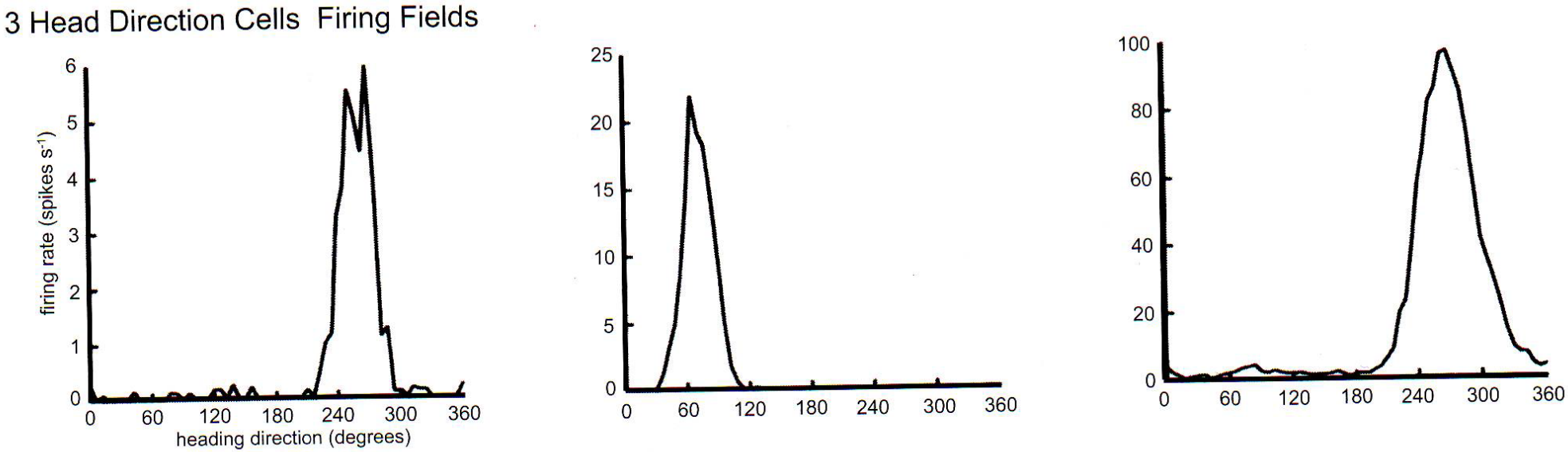
- **Maze:**
  - How do we navigate in complex environments?
  - What happens in the brain when we need to re-orient ourselves?
  - Decision making.
  - Planning.

**→ We hope to close the major gap between natural animal navigation in the wild and the neurobiology of navigation in the lab.**

# Outline of today's lecture

- Hippocampus and spatial memory: early discoveries
- Hippocampus and large-scale navigation
- Back to small-scale navigation in the laboratory:
  - Place cells
  - **Head direction cells**
  - Grid cells
  - Other brain areas involved in navigation
- Summary

# Head direction cells in dorsal presubiculum of rats




- Head direction cells are found in the dorsal presubiculum, anterior thalamus, medial entorhinal cortex, and in several other brain areas adjacent to the hippocampus.
- These cells are tuned to head direction, but *not* to place – i.e. they serve as neural “compasses”.

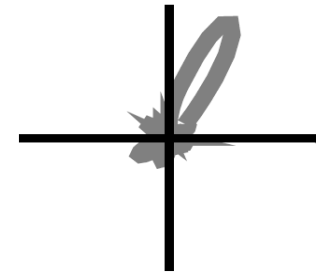
# Is there a representation of 3-D head direction in the mammalian brain = "3-D neural compasses" ?

Head-direction cells
In rats



Solstad et al.
Science 2008

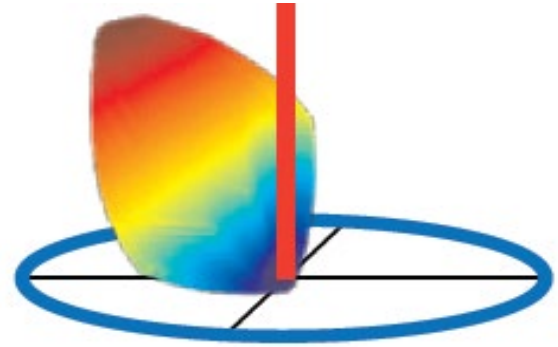
Head-direction cells
In bats



Yarsev, Witter, Ulanovsky
Nature 2011

# 3D head-direction cells in bats – a 3D neural compass

3D head-direction cell



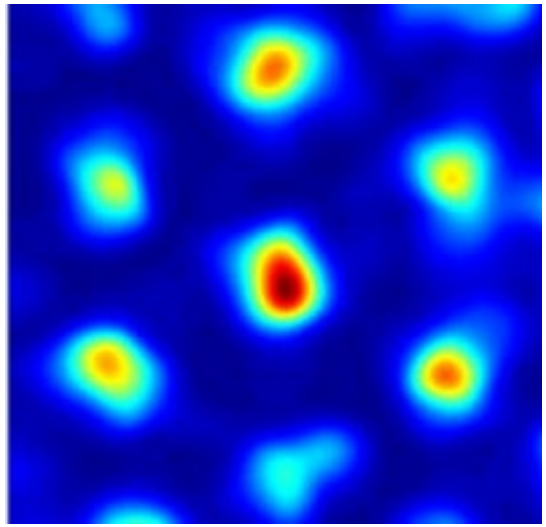
Finkelstein et al.
*Nature* (2015)

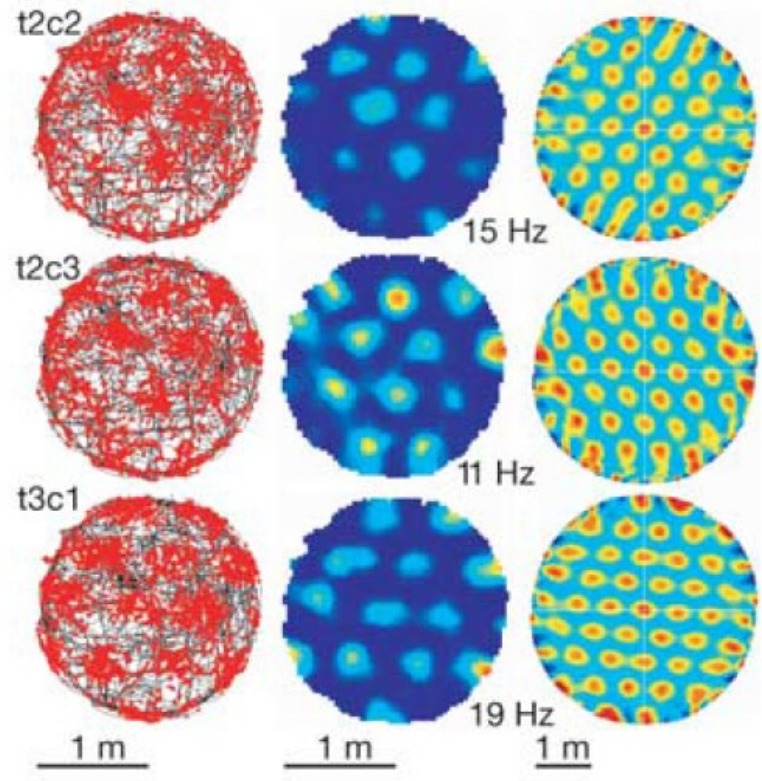
# Outline of today's lecture

- Hippocampus and spatial memory: early discoveries
- Hippocampus and large-scale navigation
- Back to small-scale navigation in the laboratory:
  - Place cells
  - Head direction cells
  - **Grid cells**
  - Other brain areas involved in navigation
- Summary

# Grid cells in medial entorhinal cortex (MEC)

Cell 1



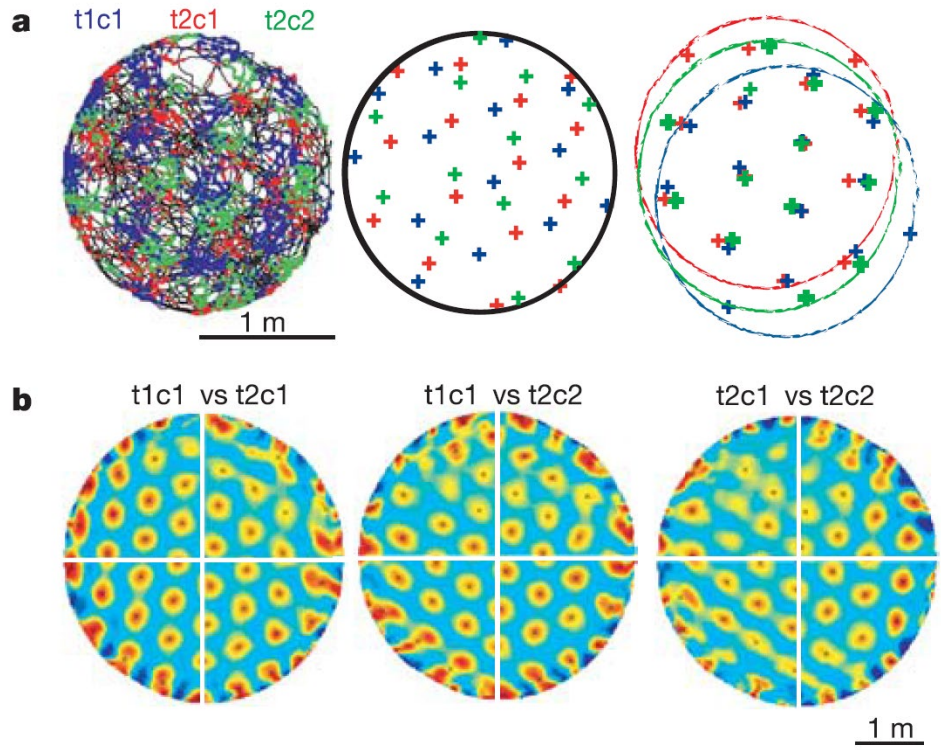



Cell 2

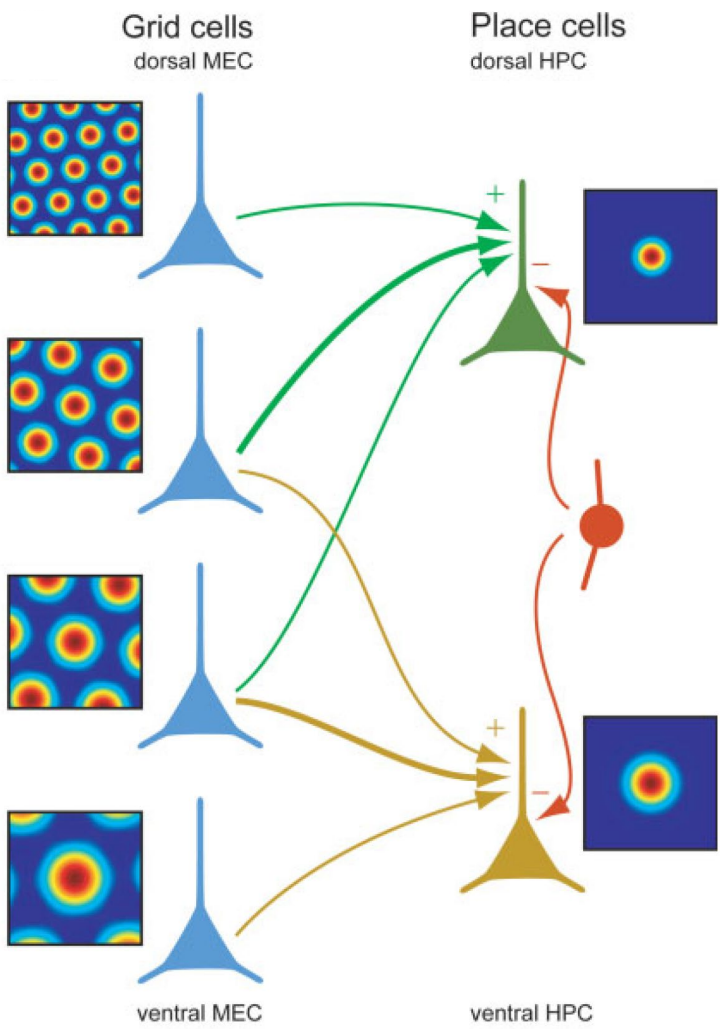
Cell 3

Cell 4

Grid cells are organized in *modules* with different spatial scales. Cells from the same module share the same grid spacing and orientation, but have a random grid phase.



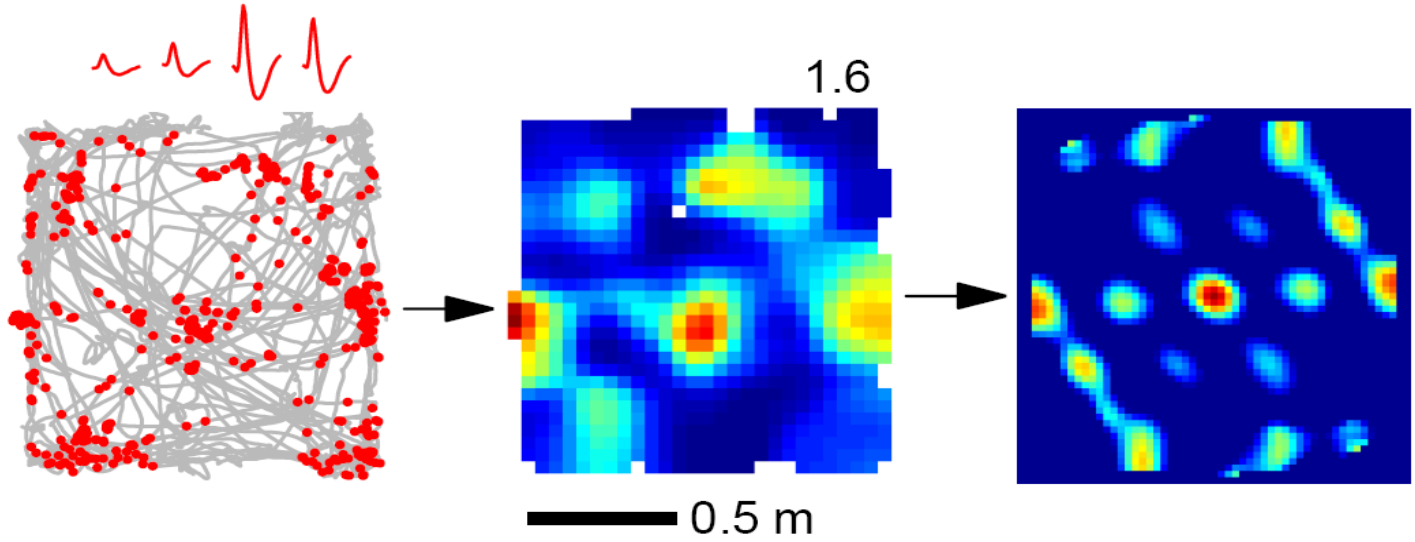
# Entorhinal grids might be combined to produce hippocampal place fields: Hexagonal Fourier-like decomposition




*Model*
by Solstad et al. (2006)

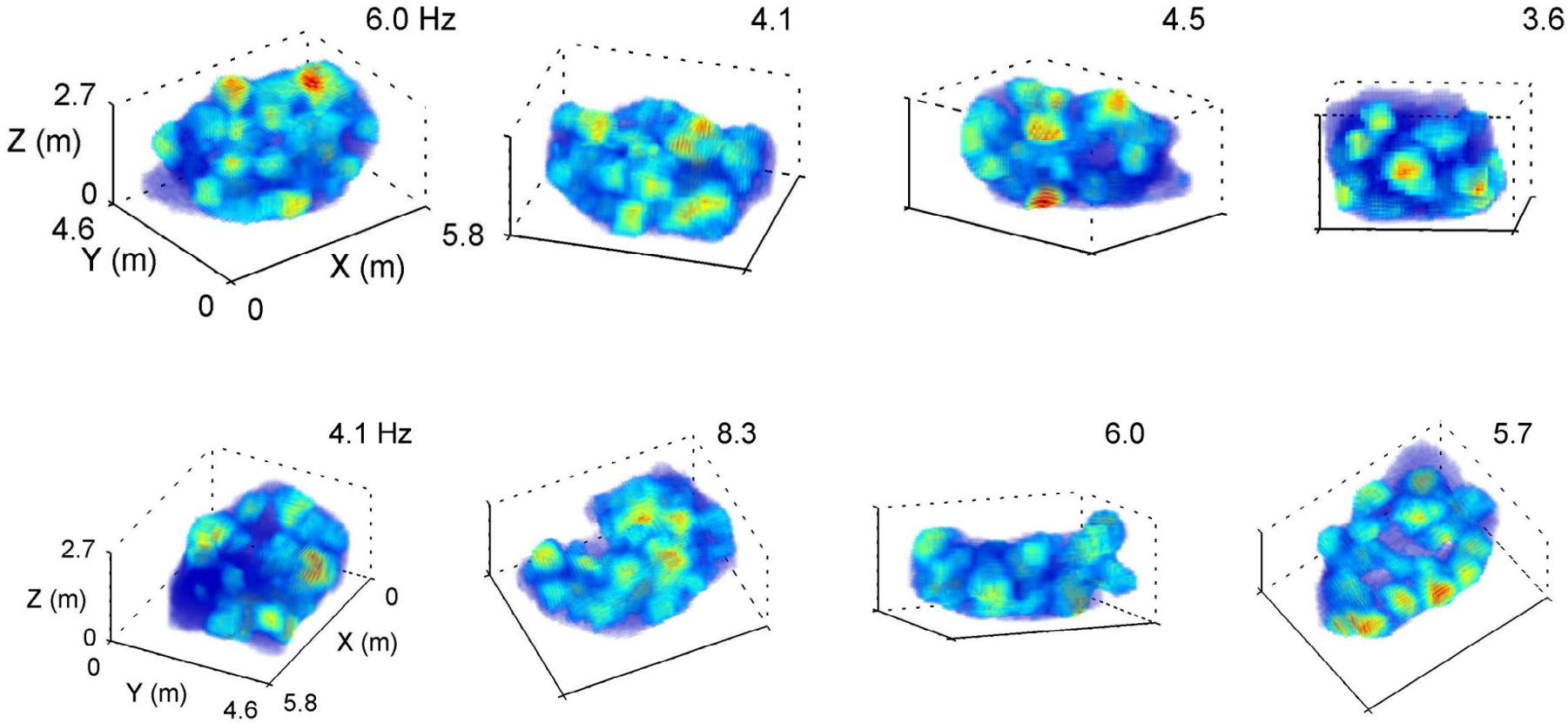
- Grid cells may provide the basis functions for the representation of space in the mammalian brain
- But: Findings in recent years have complicated this picture… Perhaps in fact the place cells generate the grid cells – or maybe it's a loop.

# Grid cells in bats – in 2D



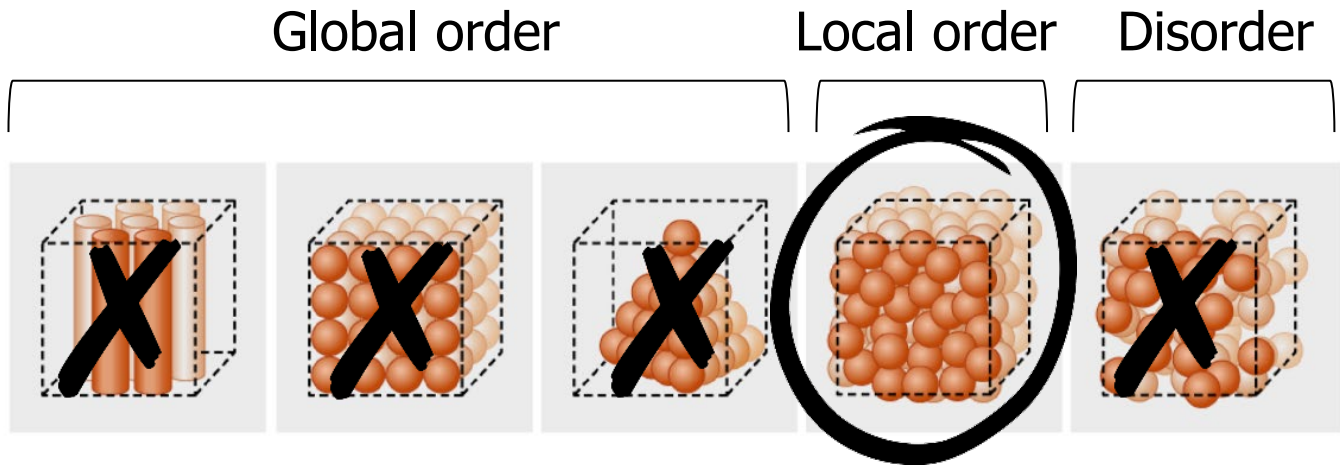

Yartsev, Witter, Ulanovsky
*Nature* (2011)

# 3D grid cells in bats



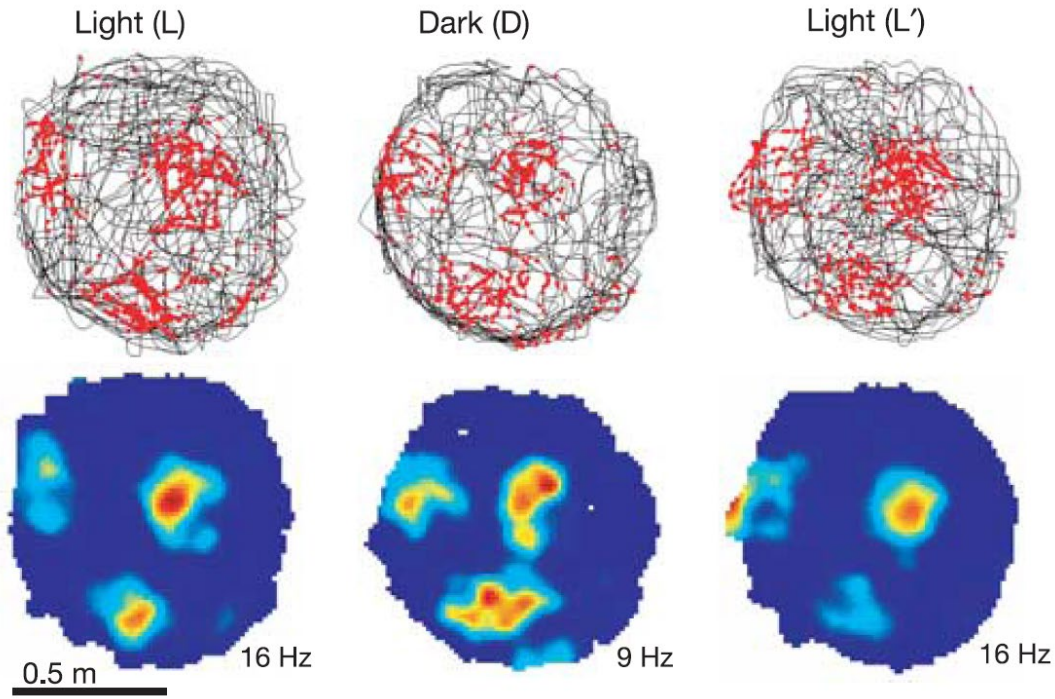

Ginosar et al.
*Nature* (2021)

# Our analyses revealed that 3D grid cells exhibit Fixed Local Distances – but No global lattice



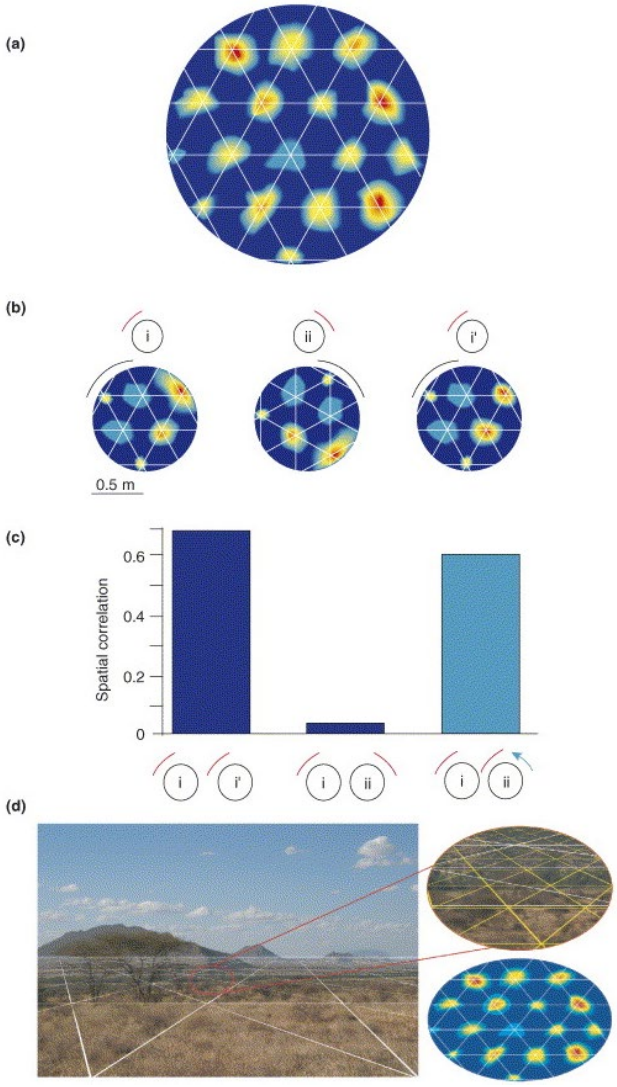

- These results have major implications for models of grid cells – both mechanistic models and functional models – which rely centrally on the global lattice structure of the grid.
- Similar results were found also in rodents (Jeffery lab).

# Proposed role of grid cells in path integration (in Rats)




- *Finding*: Grid cells persist after turning off the light (Hafting et al. 2005, see example above).
- *Caveat*: In these experiments, there was no attempt to remove odors (local cues): i.e., the rats could have been using a route-following navigational strategy (via local olfactory landmarks) to know their location – and not necessarily path integration.

# Hypothesized role of grid cells in large-scale navigation




How a researcher of bird magnetic navigation imagined grid cells might be useful for long-distance navigation (Frost & Mouritsen, *Curr. Opin. Neurobiol.* 2006)

*BUT*: No such huge grids were studied, to date.

# Question:

Neural coding in the "brain's navigation circuit" during NATURAL navigation – outdoors ?





# Latham Island, Tanzania



# Neuroscience going wild:

## Neural codes for space in bats navigating outdoors on a remote oceanic island



Shaked Palgi



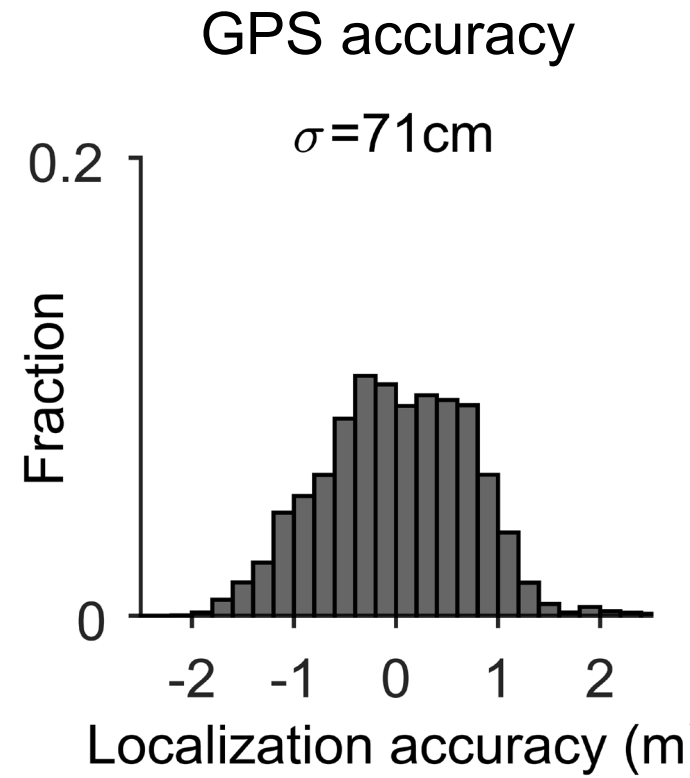
Saikat Ray

# Neuroscience Going Wild

## Adding GPS + Altimeter to the Neural Logger

Neural Logger 2024:

- 64 neural channels (16 tetrodes)
- Ultrasonic Microphone
- Accelerometer, Gyro, Magnetometer
- GPS
- Altimeter



# Neuroscience Going Wild

## Setting up a surgery room in Tanzania






# Neuroscience Going Wild



# Neuroscience Going Wild







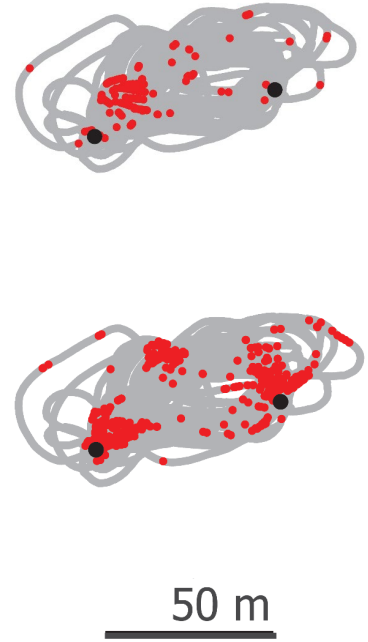






# We recorded hippocampal place cells outdoors, on the island



# We asked: are head direction cells really a neural compass ?

Head-direction cells are often called “Neural Compasses”.
But if they are indeed a Compass, there are two major requirements:

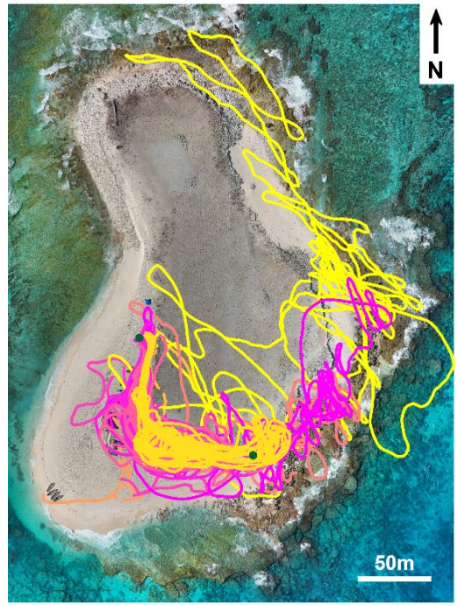
- Stability of compass over geographical space
- Stability of compass despite movement of moon & stars

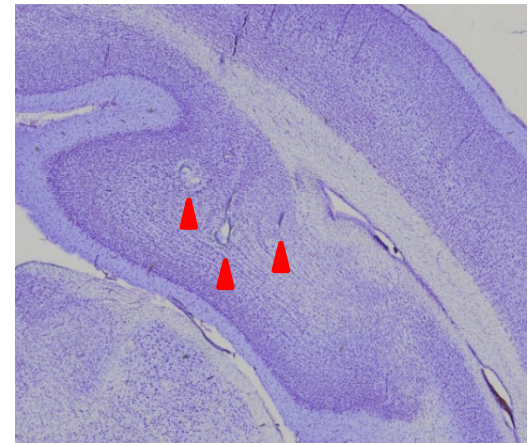
→ These two key properties were never demonstrated in the 40 years of research on head-direction cells.

→ Conundrum: laboratory experiments contradict the requirement for stability of the compass despite moon movements.

# We recorded flight-trajectories and head-direction cells on the island

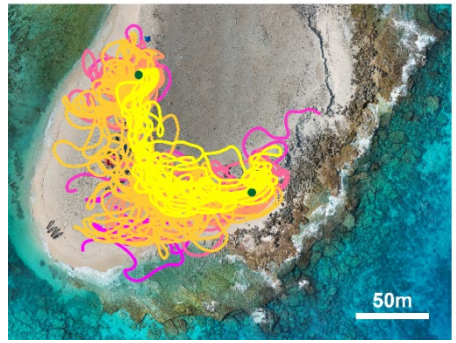
**Bat 1**

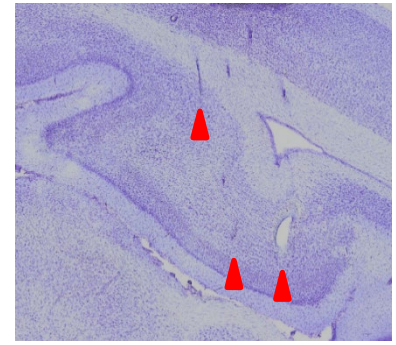


Recording sites in PrS of Bat 1

**Bat 2**



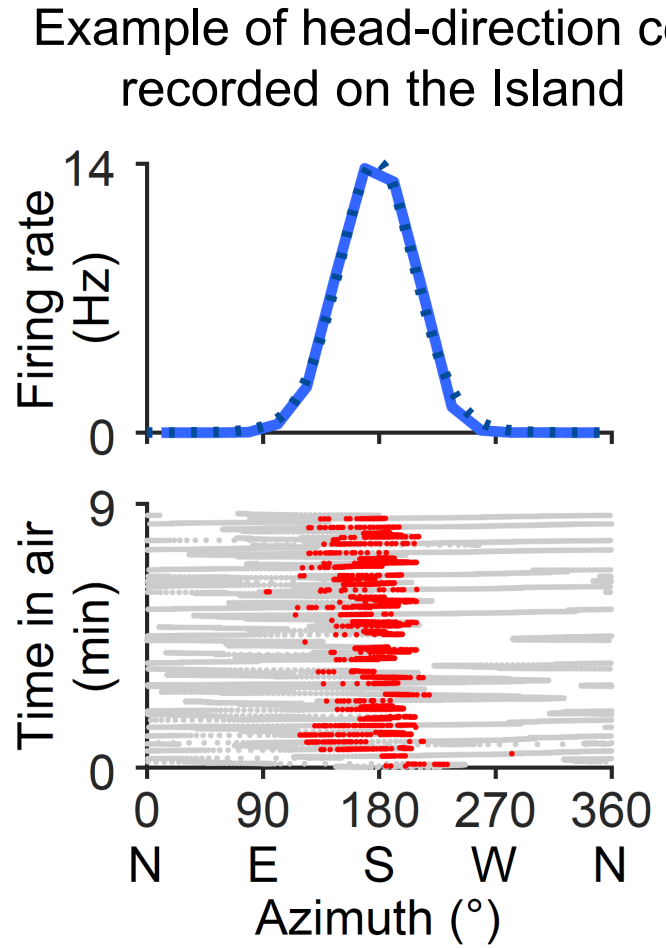



Recording sites in PrS of Bat 2

Palgi et al.
*Science* (2025)

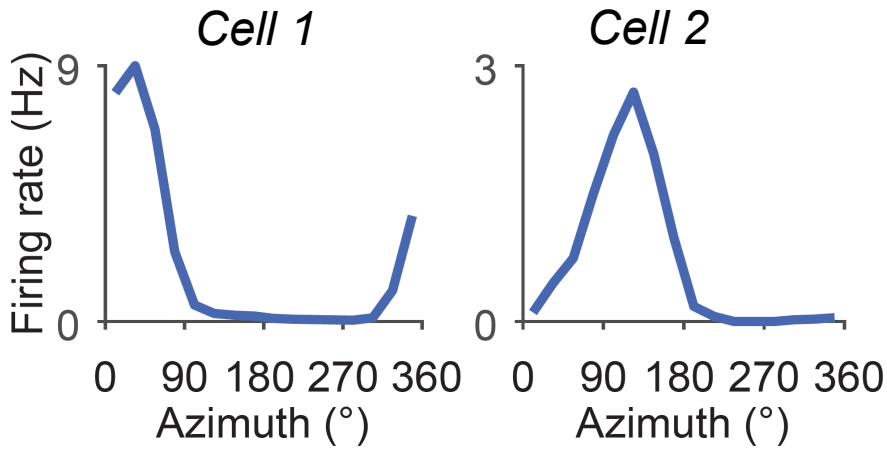
We found head direction cells outdoors !

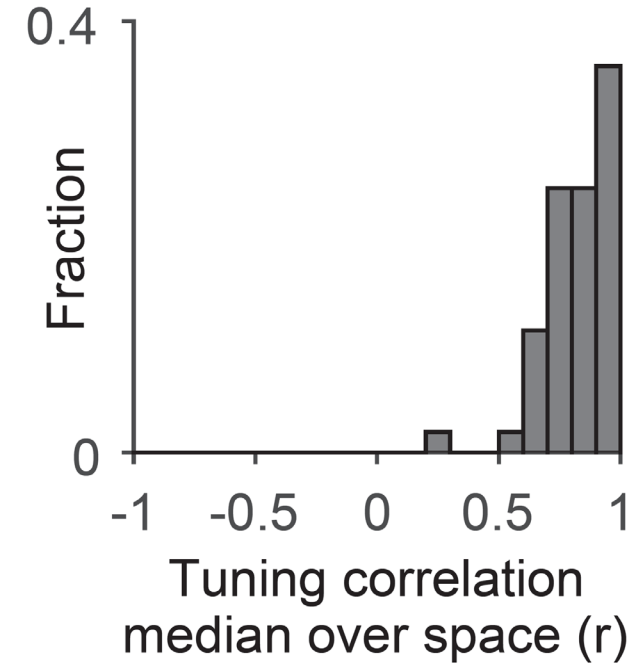
This head-direction cell points South.




Palgi et al.
*Science* (2025)

# Direction of each cell was stable over geographical space



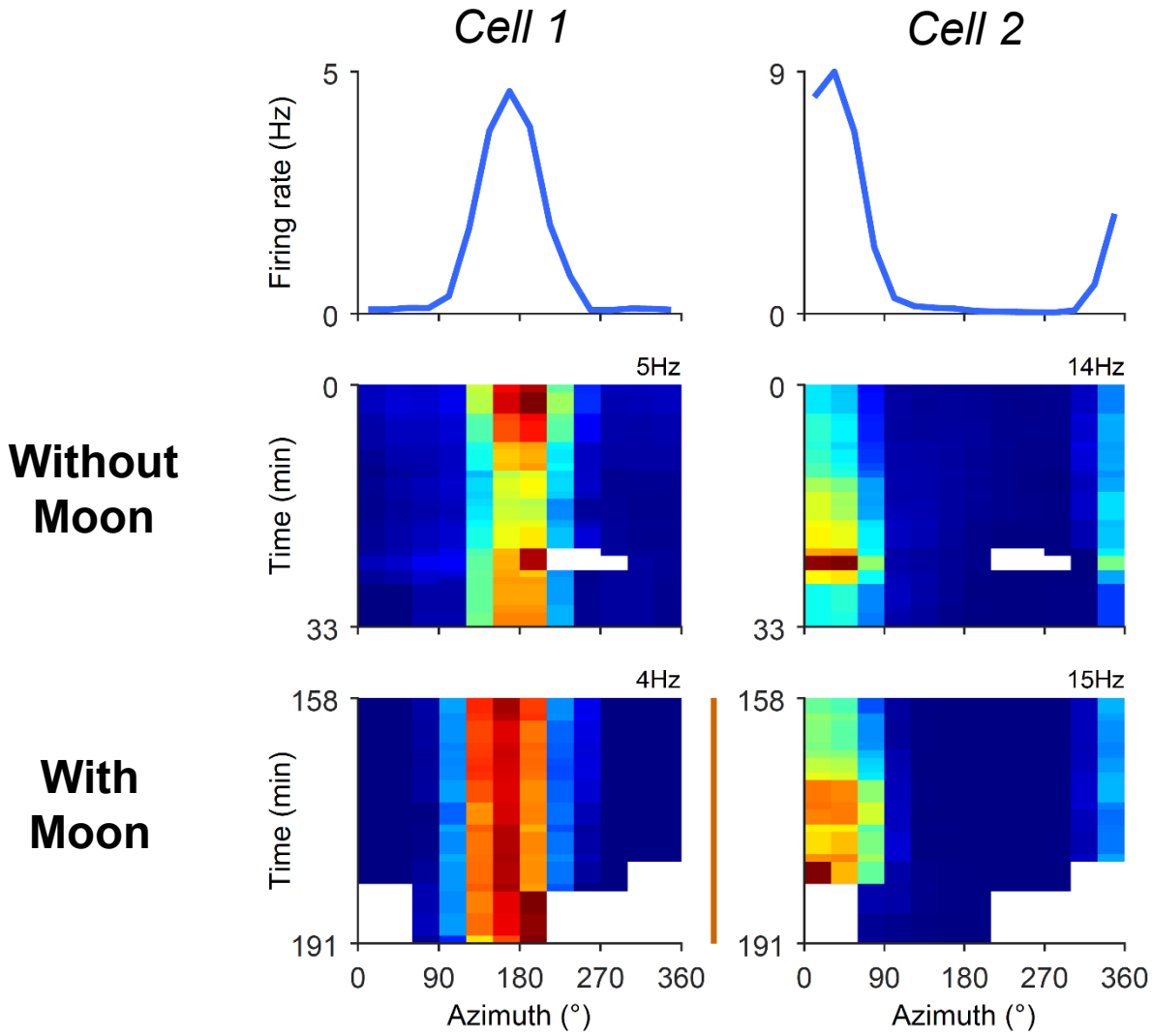







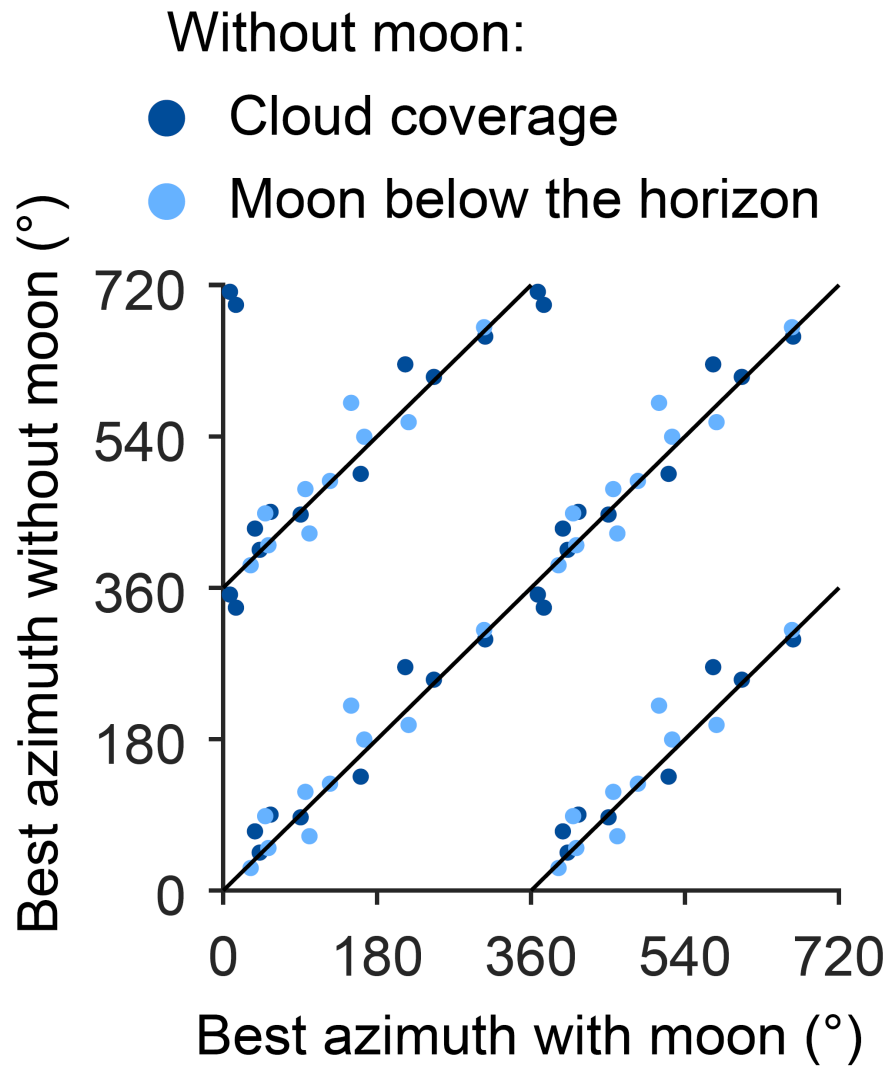
Palgi et al.
*Science* (2025)

# Direction of each cell was stable despite moon movements




Palgi et al.
*Science* (2025)

# Direction of each cell was stable despite moon movements, and stable also on overcast cloudy nights (no moon / stars)




Palgi et al.
*Science* (2025)

# These results confirm the two major requirements for a "Neural Compass" – during real navigation, outdoors

- ✓ Stability of directional tuning over geographical space
- ✓ Stability of tuning despite movement of moon & stars

→ Head direction cells *can* in fact serve as a neural compass for outdoors navigation.

# Neuroscience Going Wild

## Some Additional Questions:

- Coding of large open spaces outdoors by hippocampal place cells ?
- Coding of altitude by place cells ?
- Coding of sea/land borders by border cells in the subiculum ?
- Representation of goals during natural goal-directed navigation outdoors ?
- Grid cells in the real world ?
- Replay of natural navigational trajectories during sleep ?
- etc, etc, etc…

**→ These are the first-ever neural recordings in mammals outdoors – in the brain of any mammalian species – so anything we shall find will be extremely interesting and exciting !**

# Outline of today's lecture

- Hippocampus and spatial memory: early discoveries
- Hippocampus and large-scale navigation
- Back to small-scale navigation in the laboratory:
  - Place cells
  - Head direction cells
  - Grid cells
  - **Other brain areas involved in navigation**
- Summary

# Other brain areas involved in navigation

* Beaconing / "response strategy" – **striatum**. That is, if you train a rat to always turn left (response strategy): this depends on the striatum. But if you train the rat to reach some *absolute* location in space: this depends on intact hippocampus & entorhinal cortex.

* Transformations from organism-based ("egocentric") coordinate frame, in which sensory information is acquired, to absolute-space-based ("allocentric") coordinate frame – **parietal cortex** ?

* Transformation from absolute-space-based ("allocentric") coordinate frame to the motor action of navigation – **Retrosplenial cortex** ?

* High cognitive functions in navigation (planning, decision-making, optimization) – **Prefrontal cortex** ?

→ Very little is known about these more complex aspects of navigation !

# Outline of today's lecture

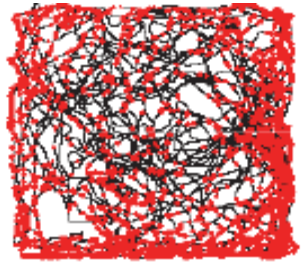
- Introduction: Feats of animal navigation
- Navigational strategies:
  - Beaconing
  - Route following
  - Path integration
  - Map and Compass / Cognitive Map
- Sensory cues for navigation:
  - Compass mechanisms
  - Map mechanisms
- Brain mechanisms of Navigation (brief introduction)
- **Summary**

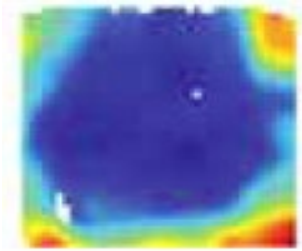
# Spatial cell types in the hippocampus and entorhinal cortex: The basic elements of the rat's "brain navigation circuit"

Medial entorhinal cortex

Hippocampus

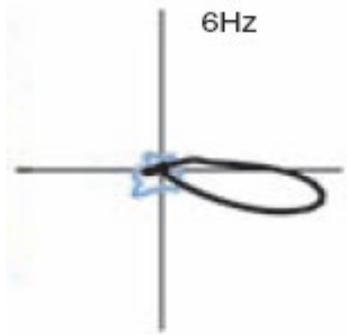
Border Cells





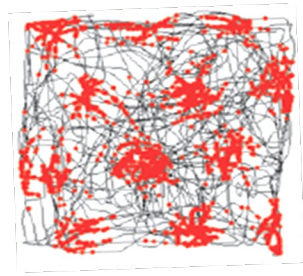
Mosers, Knierim, O'Keefe

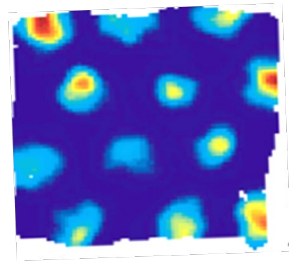
Head-direction cells

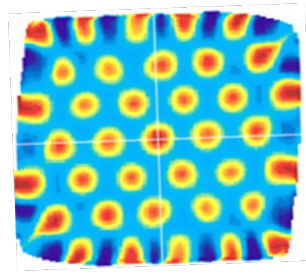



Ranck, Taube

Grid cells

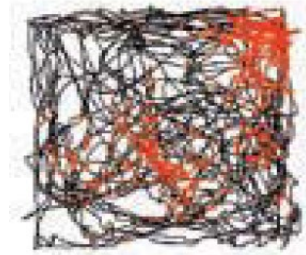


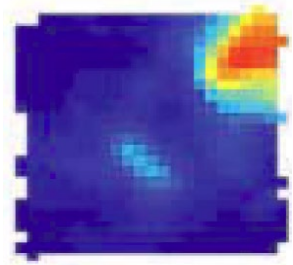




Mosers

Place cells





O'Keefe

# Spatial cell types in the hippocampus and entorhinal cortex: The basic elements of the rat's "brain navigation circuit"

**SUMMARY:**

- **Place cell** → Position (where am I)
- **Grid cells** → Distance ("ruler")
- **Border cells** → Borders of the environment

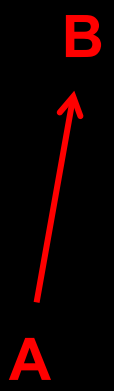
} Map

- **Head-direction cells** → Direction

} Compass

# What's missing? How to navigate to goals !

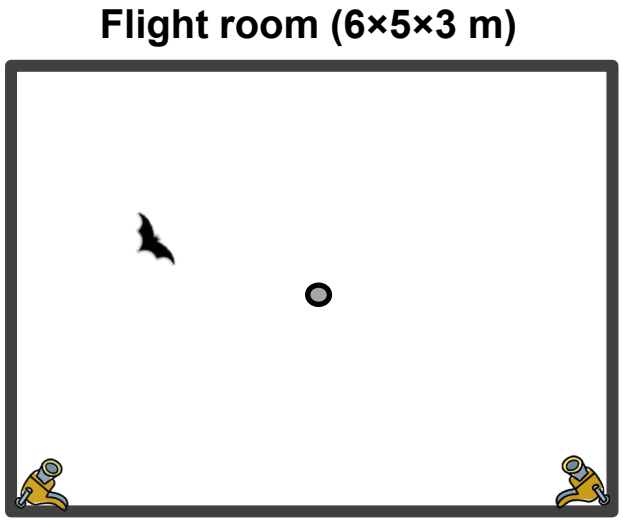
## Vectorial representation of navigational goals in the bat hippocampus




Ayelet Sarel

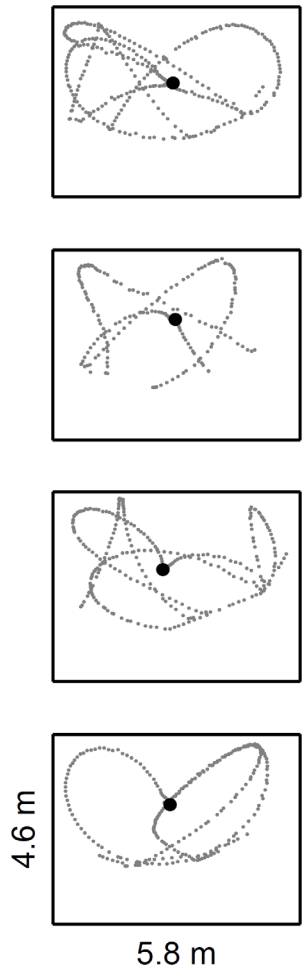


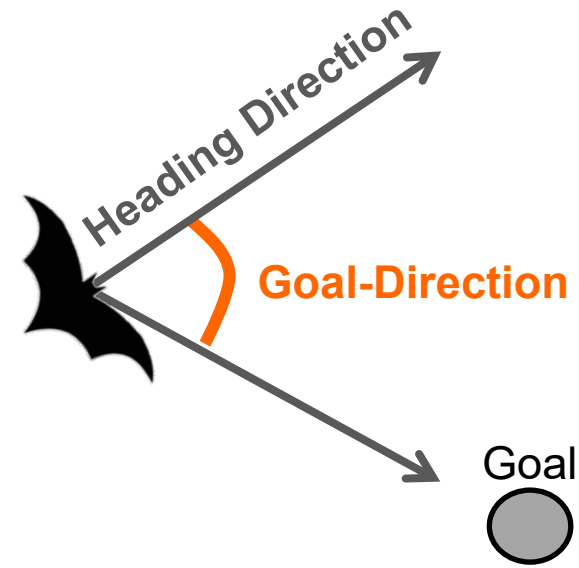
# A goal-directed behavioral task in bats

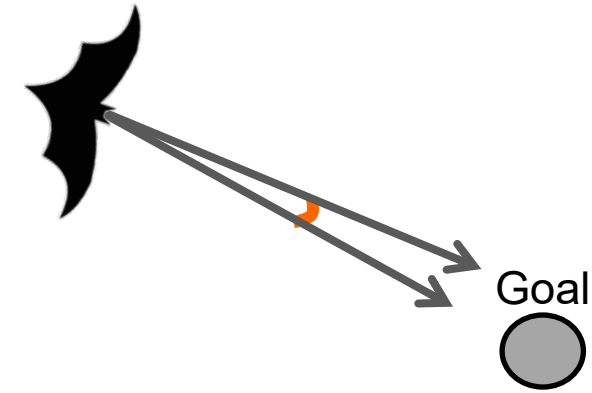





Recordings in hippocampal area CA1



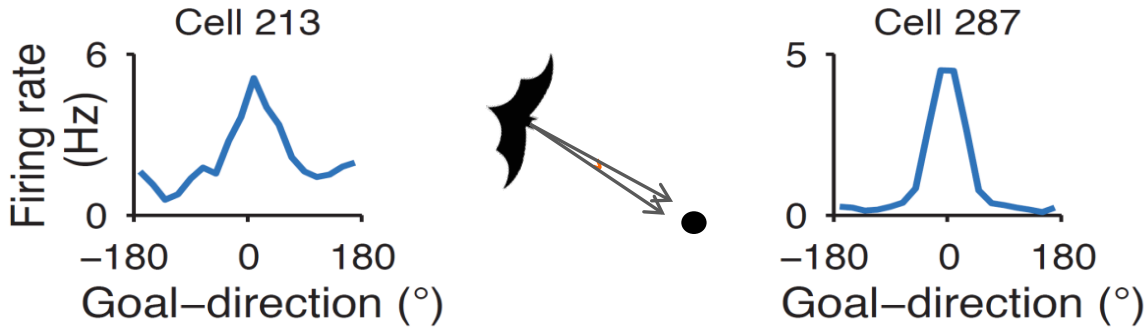



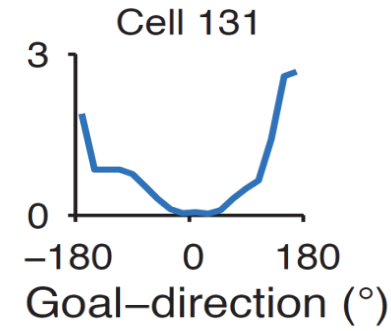

Sarel et al., *Science* (2017)

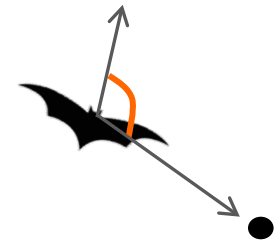
# Goal-Direction cells: hippocampal CA1 cells with tuning to the goal's direction

**Zero preferred angle**




**Non-zero preferred angle**



- These neurons represented direction also to an occluded goal → a memory-based representation.
- Many of these neurons encoded also the distance-to-goal → egocentric vectorial coding of goals.
- In the Maze we plan to study how *multiple* goals are represented by goal-direction cells.

Sarel et al., *Science* (2017)

# Future challenges in the study of the neural basis of animal navigation, spatial memory and spatial cognition

- **Gap in spatial scale**: Even rats (let alone bats) would require in the wild neural codes for MUCH larger spaces than shown to date in the laboratory.
  → Our experiments in the Tunnel and on the Island start closing this gap.

- **Navigation in Complex environments** ?!
  → Our Maze experiments will hopefully start closing this gap.

- **Too little is known about the neural basis of the "higher" components of navigation** (apart of the "map" & "compass" components) – and this needs to be elucidated:
  - Trajectory planning ?
  - Decision making ?
  - Re-orientation ?

